

Complex Analysis

Module 3CA

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Chapter 0

The complex numbers

You must be familiar with complex numbers from your first year courses. In this chapter we give a short overview of the most important concepts.

Formally, we define the set of *complex numbers* $\mathbb{C}$ as the set of ordered pairs of reals $z = (x, y)$, $x, y \in \mathbb{R}$, with the operations of addition and multiplication defined as follows:

$$(x, y) + (u, v) = (x + u, y + v),$$
$$(x, y) \cdot (u, v) = (xu - yv, xv + yu).$$

The number $x = \operatorname{Re} z$ is called the *real part* of z , and the number $y = \operatorname{Im} z$ is called the *imaginary part* of z .

A real number $x \in \mathbb{R}$ is identified with the complex number $(x, 0) \in \mathbb{C}$. In this way we obtain the inclusion $\mathbb{R} \subseteq \mathbb{C}$. We also introduce a special number $i = (0, 1)$ which is called the *imaginary unit*. Directly from the definition of the multiplication we see that

$$i^2 = -1.$$

Now it is convenient to write complex numbers in the form

$$z = x + iy.$$

One can easily see that this is consistent with the definitions of addition and multiplication.

The complex number $z = (x, y)$ can be geometrically represented as the point with the Cartesian coordinates (x, y) in a plane (see [Figure 0.1](#)). The plane formed by complex numbers is called the *complex plane*. The x -axis is formed by the real numbers and is called the *real axis*. The y -axis is called the *imaginary axis*; it is formed by the numbers of the form iy , $y \in \mathbb{R}$, that are called the *imaginary numbers*.

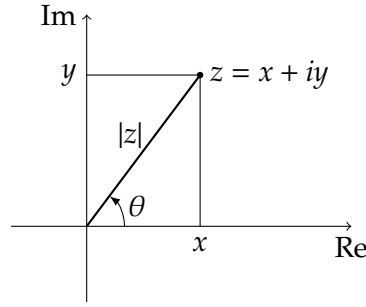


Figure 0.1: The complex plane

The *modulus* of the complex number $z = x + iy$ is

$$|z| = \sqrt{x^2 + y^2}.$$

If z is real, then $|z|$ is just the absolute value of z . Geometrically, $|z|$ is the distance from the point z to the origin in the complex plane. Moreover, $d(z, w) = |z - w|$, $z, w \in \mathbb{C}$, is a metric on $\mathbb{C}$; the value $|z - w|$ is exactly the distance between the points z and w in the plane.

We can represent the number $z = x + iy$ also in polar coordinates (r, θ) , where $r = |z| = \sqrt{x^2 + y^2}$ and, for $z \neq 0$, θ is the angle between the positive real axis and the segment connecting the point z and the origin. We have $\tan \theta = \frac{y}{x}$. We call θ the *argument* of the complex number z and write $\theta = \arg z$. The argument of $z \neq 0$ is not uniquely determined: if θ is an argument of z , then $\theta + 2\pi n$ with any $n \in \mathbb{Z}$ is also an argument of z . Notice that $x = |z| \cos \theta$ and $y = |z| \sin \theta$. Therefore the number z can be written as

$$z = |z| (\cos \theta + i \sin \theta).$$

The argument of $z = 0$ is not defined, but an arbitrary angle θ would work. Using the exponential function, we can write

$$z = |z| (\cos \theta + i \sin \theta) = |z| e^{i\theta}.$$

We will return to the formula $e^{i\theta} = \cos \theta + i \sin \theta$ later in [Section 1.6](#) and justify it.

The operations of addition and multiplication on $\mathbb{C}$ satisfy the commutative,

associative and distributive laws, i.e., for $z_1, z_2, z_3 \in \mathbb{C}$ we have

$$\begin{aligned} z_1 + z_2 &= z_2 + z_1, & z_1 z_2 &= z_2 z_1, \\ z_1 + (z_2 + z_3) &= (z_1 + z_2) + z_3, & z_1(z_2 z_3) &= (z_1 z_2) z_3, \\ z_1(z_2 + z_3) &= z_1 z_2 + z_1 z_3. \end{aligned}$$

The number 0 is the additive identity and the number 1 is the multiplicative identity, i.e., for all $z \in \mathbb{C}$

$$0 + z = z, \quad 1z = z.$$

Moreover, for any $z = x + iy$ there exists an additive inverse $-z = (-x) + i(-y)$ such that $z + (-z) = 0$, and for any $z \neq 0$ there exists a multiplicative inverse $\frac{1}{z} = \frac{x}{|z|^2} + i\frac{-y}{|z|^2}$. These rules express the fact that $\mathbb{C}$ is a *field*. The set of real numbers $\mathbb{R}$ is a subfield of $\mathbb{C}$.

Notice that while it is easier to perform addition of complex numbers in Cartesian coordinates,

$$(x + iy) + (u + iv) = (x + u) + i(y + v),$$

the multiplication is easier in polar coordinates: if $z = |z|(\cos \theta + i \sin \theta) = |z|e^{i\theta}$ and $w = |w|(\cos \phi + i \sin \phi) = |w|e^{i\phi}$, then

$$zw = |z||w|(\cos(\theta + \phi) + i \sin(\theta + \phi)) = |z||w|e^{i(\theta + \phi)}.$$

This follows from the addition theorems for sine and cosine. A simple proof by induction gives *de Moivre's Theorem*

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta, \quad \theta \in \mathbb{R}, n \in \mathbb{N}.$$

The set $\mathbb{C}$ is equipped with one more operation: the conjugation. Given $z = x + iy$, the *complex conjugate* of z is defined by

$$\bar{z} = x - iy.$$

Geometrically, conjugation is the reflection in the real axis (see [Figure 0.2](#)). In the polar coordinates we have

$$\bar{z} = |z|(\cos \theta - i \sin \theta) = |z|e^{-i\theta}.$$

There is a number of identities involving the conjugate and the modulus of z . They can be proven quite easily from the definitions. We list some of them

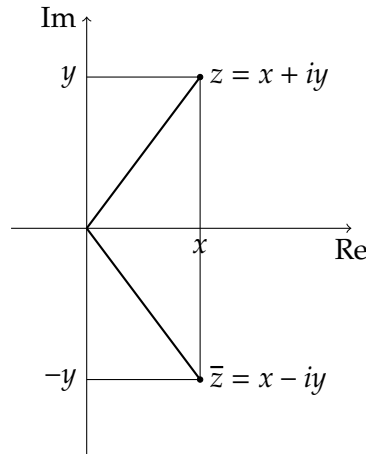


Figure 0.2: A number z and its conjugate $\bar{z}$

below: for $z, w \in \mathbb{C}$

$$\operatorname{Re} \bar{z} = \operatorname{Re} z, \quad \operatorname{Im} \bar{z} = -\operatorname{Im} z,$$

$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \quad \operatorname{Im} z = \frac{z - \bar{z}}{2i},$$

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z}\bar{w},$$

$$\bar{\bar{z}} = z, \quad |\bar{z}| = |z|, \quad z\bar{z} = |z|^2, \quad |z||w| = |zw|, \quad \frac{1}{z} = \frac{\bar{z}}{|z|^2}.$$

Very important is also the *triangle inequality*

$$|z + w| \leq |z| + |w|.$$

Further simple inequalities are

$$|\operatorname{Re} z| \leq |z|, \quad |\operatorname{Im} z| \leq |z|.$$

At this point we would like to note that, unlike for real numbers, there is no order relation on $\mathbb{C}$. No inequalities are possible between complex numbers. Notice that the inequalities above involve real numbers.

We mentioned above that $d(z, w) = |z - w|$, $z, w \in \mathbb{C}$, is a metric on $\mathbb{C}$. Thus, we can consider $\mathbb{C}$ as a metric space and introduce the corresponding topological notions like open and closed sets, and convergence.

The *open disc* with center $a \in \mathbb{C}$ and radius $\varepsilon > 0$ is the set

$$D(a; \varepsilon) = \{z \in \mathbb{C} : |z - a| < \varepsilon\}.$$

A set $S \subseteq \mathbb{C}$ is called *open*, if for any $z \in S$ there is $\varepsilon > 0$ such that $D(z; \varepsilon) \subseteq S$.

A set $S \subseteq \mathbb{C}$ is called *closed*, if its complement $\mathbb{C} \setminus S$ is open.

A set $S \subseteq \mathbb{C}$ is called *bounded*, if there is $M > 0$ such that $|z| \leq M$ for all $z \in S$.

A point $a \in \mathbb{C}$ is called a *limit point* of a set $S \subseteq \mathbb{C}$, if for any $\varepsilon > 0$ the set $D(a; \varepsilon) \setminus \{a\}$ intersects S .

Definition 0.1 We say that a complex sequence $(z_n)_{n=1}^\infty$ *converges* to a *limit* $a \in \mathbb{C}$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |z_n - a| < \varepsilon.$$

In this case we write $a = \lim_{n \rightarrow \infty} z_n$, or $z_n \rightarrow a$ as $n \rightarrow \infty$.

△

Lemma 0.2 A complex sequence $(z_n)_{n=1}^\infty$ converges if and only if the both real sequences $(\operatorname{Re} z_n)_{n=1}^\infty$ and $(\operatorname{Im} z_n)_{n=1}^\infty$ converge. Moreover, if $z_n \rightarrow a$, then $\operatorname{Re} z_n \rightarrow \operatorname{Re} a$ and $\operatorname{Im} z_n \rightarrow \operatorname{Im} a$, $n \rightarrow \infty$.

△

Proof. First assume that $\operatorname{Re} z_n \rightarrow u$ and $\operatorname{Im} z_n \rightarrow v$, $n \rightarrow \infty$, with some $u, v \in \mathbb{R}$. Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$|\operatorname{Re} z_n - u| < \frac{\varepsilon}{\sqrt{2}}, \quad |\operatorname{Im} z_n - v| < \frac{\varepsilon}{\sqrt{2}}, \quad n \geq N.$$

Then for $n \geq N$

$$|z_n - (u + iv)| = \sqrt{(\operatorname{Re} z_n - u)^2 + (\operatorname{Im} z_n - v)^2} < \sqrt{\frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{2}} = \varepsilon.$$

Thus, $z_n \rightarrow u + iv$, $n \rightarrow \infty$.

For the opposite direction, suppose that $z_n \rightarrow a$, $n \rightarrow \infty$. Then given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$

$$|z_n - a| < \varepsilon.$$

But this implies

$$|\operatorname{Re} z_n - \operatorname{Re} a| \leq |z_n - a| < \varepsilon, \quad |\operatorname{Im} z_n - \operatorname{Im} a| \leq |z_n - a| < \varepsilon, \quad n \geq N.$$

Thus, $\operatorname{Re} z_n \rightarrow \operatorname{Re} a$ and $\operatorname{Im} z_n \rightarrow \operatorname{Im} a$, $n \rightarrow \infty$.

□

The lemma above states that convergence of a complex sequence $(z_n)_{n=1}^{\infty}$ is equivalent to convergence of the two real sequences $(\operatorname{Re} z_n)_{n=1}^{\infty}$ and $(\operatorname{Im} z_n)_{n=1}^{\infty}$. This allows us to easily transfer many theorems about limits of real sequences to the complex case like the algebraic limit theorems and the following statements. We will not go into details.

Any bounded sequence in $\mathbb{C}$ has a convergent subsequence.

A set $K \subseteq \mathbb{C}$ is compact (i.e., any sequence in K has a subsequence that converges to an element of K) if and only if it is closed and bounded.

The space $\mathbb{C}$ is complete (i.e., every Cauchy sequence in $\mathbb{C}$ converges in $\mathbb{C}$).

Chapter 1

Holomorphic functions

1.1 Derivatives

Let $G \subseteq \mathbb{C}$ be a nonempty set. Mostly we will consider open sets G . We will study functions $f : G \rightarrow \mathbb{C}$.

The definitions of a limit of a function, of a continuous function and of a derivative of a function for complex functions are practically identical to the definitions in the real case.

Definition 1.1 Let $f : G \rightarrow \mathbb{C}$, where $G \subseteq \mathbb{C}$ is nonempty, and let a be a limit point of G . We say that f *converges* to a *limit* $w \in \mathbb{C}$ as $z \rightarrow a$ if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall z \in G : 0 < |z - a| < \delta \implies |f(z) - w| < \varepsilon.$$

We write $\lim_{z \rightarrow a} f(z) = w$ or, alternatively, $f(z) \rightarrow w$ as $z \rightarrow a$.

△

For $f : G \rightarrow \mathbb{C}$ we can write

$$f = \operatorname{Re} f + i \operatorname{Im} f,$$

where $\operatorname{Re} f, \operatorname{Im} f : G \rightarrow \mathbb{R}$ are real-valued functions given by $(\operatorname{Re} f)(z) = \operatorname{Re} f(z)$, $(\operatorname{Im} f)(z) = \operatorname{Im} f(z)$. Exactly as for sequences, we have the following statement.

Proposition 1.2 $f(z) \rightarrow w$ if and only if $\operatorname{Re} f(z) \rightarrow \operatorname{Re} w$ and $\operatorname{Im} f(z) \rightarrow \operatorname{Im} w$ as $z \rightarrow a$.

△

We leave the proof to the reader (modify the proof of [Lemma 0.2](#)).

Definition 1.3 Let $f : G \rightarrow \mathbb{C}$, where $S \subseteq \mathbb{C}$ is an open nonempty set. We say that f is *continuous* at a point $a \in G$ if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall z \in G : |z - a| < \delta \implies |f(z) - f(a)| < \varepsilon.$$

We say that f is *continuous on G* if f is continuous at each point of G .

△

The algebraic properties of limits and of continuous functions can be developed exactly as in the real case. We do not repeat them here.

Definition 1.4 Let $f : G \rightarrow \mathbb{C}$ where G is an open nonempty set in $\mathbb{C}$. We say that f is *differentiable* at a point $z \in G$ if the limit

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

exists. If this is the case, we call the value

$$f'(z) = \frac{df}{dz}(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

the *derivative* of f at z .

△

Notice that h in the definition above is a complex number, so that $z + h$ approaches z along any path. The value of the limit does not depend on the manner in which $z + h$ approaches z .

As in the real case, except in simple examples it is difficult to check using the definition whether a given function is differentiable at a given point. However, as **Definition 1.4** is a verbatim repetition of the definition of a derivative for real functions, also the rules for differentiation such as the sum, the product, the quotient and the chain rules, can be transferred to the complex derivatives. The proofs repeat the proofs in the real case.

Also the next property is the exact repetition of the corresponding property in the real case.

Proposition 1.5 The function $f : G \rightarrow \mathbb{C}$ (where G is an open nonempty set in $\mathbb{C}$) is differentiable at $z \in G$ if and only if there exist a number $m \in \mathbb{C}$ and a complex function $\epsilon(h)$ with $\lim_{h \rightarrow 0} \epsilon(h) = 0$ such that for all $h \in \mathbb{C}$ with $z + h \in G$ we have

$$f(z+h) = f(z) + mh + \epsilon(h)h.$$

△

Proof. If the above representation holds, then

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = \lim_{h \rightarrow 0} \frac{mh + \epsilon(h)h}{h} = \lim_{h \rightarrow 0} (m + \epsilon(h)) = m,$$

so that f is differentiable at z and $f'(z) = m$.

On the other hand, if f is differentiable at z , then we can take $m = f'(z)$ and

$$\epsilon(h) = \begin{cases} \frac{f(z+h)-f(z)}{h} - f'(z), & h \neq 0, \\ 0, & h = 0, \end{cases}$$

where the function ϵ above is defined for all $h \in \mathbb{C}$ with $z+h \in G$. Notice that G is open, and thus there is a small $\rho > 0$ such that $D(z; \rho) \subseteq G$. From the definition of the derivative $f'(z)$ it follows that $\lim_{h \rightarrow 0} \epsilon(h) = 0$. □

A straightforward implication of this result is the following statement.

Corollary 1.6 *If f is differentiable at $z \in G$, then f is continuous at z .* △

As in the real case, the converse statement does not hold.

Example 1.7 Consider the function $f(z) = |z|$.

From the inequality $\|z\| - \|w\| \leq \|z - w\|$, $z, w \in \mathbb{C}$ it follows that $f(z) = |z|$ is continuous on $\mathbb{C}$. Indeed, given $\varepsilon > 0$ we take $\delta = \varepsilon > 0$, then

$$\|z - w\| < \varepsilon \implies \|z\| - \|w\| < \varepsilon.$$

However, $f(z) = |z|$ is not differentiable at $z = 0$. Indeed, if we take h real and positive in the limit in the definition of the derivative, we obtain

$$\lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}, h > 0}} \frac{f(h) - f(0)}{h} = \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}, h > 0}} \frac{h}{h} = 1,$$

but if we take h real and negative, we have

$$\lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}, h < 0}} \frac{f(h) - f(0)}{h} = \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}, h < 0}} \frac{-h}{h} = -1.$$

Thus, the limit

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

does not exist. △

We are now ready to give one of the most important definitions in this course.

Definition 1.8 (1) A function f is called *holomorphic* in an open set G if it is differentiable at each point $z \in G$.

(2) Let $S \subseteq \mathbb{C}$ be an arbitrary set in $\mathbb{C}$. We say that f is *holomorphic* in S , if there exists an open set G such that $S \subseteq G$ and f is holomorphic in G . $\triangle$

Example 1.9 Any polynomial

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0,$$

where $a_n, a_{n-1}, \dots, a_0 \in \mathbb{C}$, is holomorphic in $\mathbb{C}$.

Any rational function $\frac{p}{q}$, where p and q are complex polynomials, is holomorphic in its domain. $\triangle$

1.2 The Cauchy-Riemann equations

Let $f : G \rightarrow \mathbb{C}$ be a function defined on an open set $G \subseteq \mathbb{C}$. Writing $z = x + iy$ and $f(z) = \operatorname{Re} f(z) + i \operatorname{Im} f(z)$, we see that we can represent f in the form

$$f(z) = u(x, y) + iv(x, y),$$

where u and v are real-valued functions defined for $(x, y) \in G$ which is now considered as a subset of $\mathbb{R}^2$.

For a real-valued function u of two variables, we denote its partial derivatives of the first order by $\frac{\partial u}{\partial x} = u_x$, $\frac{\partial u}{\partial y} = u_y$.

Theorem 1.10 (The Cauchy-Riemann Equations) Let $f : G \rightarrow \mathbb{C}$ be a function defined on an open set $G \subseteq \mathbb{C}$. If f is differentiable at a point $z = x + iy \in G$, then the functions u and v have partial derivatives of first order at the point (x, y) , and

$$u_x(x, y) = v_y(x, y), \quad u_y(x, y) = -v_x(x, y).$$

$\triangle$

Proof. Recall that

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}.$$

Chapter 1 Holomorphic functions

The limit of the quotients $\frac{f(z+h)-f(z)}{h}$ is independent of the path through which $h \in \mathbb{C}$ approaches the origin. Let us take $h \in \mathbb{R}, h \rightarrow 0$. We obtain

$$\begin{aligned} f'(z) &= \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{f(z+h) - f(z)}{h} \\ &= \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{[u(x+h, y) + iv(x+h, y)] - [u(x, y) + iv(x, y)]}{h} \\ &= \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{u(x+h, y) - u(x, y)}{h} + i \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{v(x+h, y) - v(x, y)}{h} \\ &= u_x(x, y) + iv_x(x, y). \end{aligned}$$

The existence of the limits $\lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{u(x+h, y) - u(x, y)}{h} = u_x(x, y)$ and $\lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{v(x+h, y) - v(x, y)}{h} = v_x(x, y)$ follows from **Proposition 1.2**.

Now let us consider the case when h approaches zero along the imaginary axis, i.e. $h = ik, k \in \mathbb{R}, k \rightarrow 0$. Similarly to the above,

$$\begin{aligned} f'(z) &= \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{f(z+ik) - f(z)}{ik} \\ &= \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{[u(x, y+k) + iv(x, y+k)] - [u(x, y) + iv(x, y)]}{ik} \\ &= \frac{1}{i} \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{u(x, y+k) - u(x, y)}{k} + \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{v(x, y+k) - v(x, y)}{k} \\ &= -iu_y(x, y) + v_y(x, y). \end{aligned}$$

Also here we obtain that the partial derivatives $u_y(x, y), v_y(x, y)$ exist.

We see that

$$f'(z) = u_x(x, y) + iv_x(x, y) = -iu_y(x, y) + v_y(x, y).$$

Equating real and imaginary parts, we obtain

$$u_x(x, y) = v_y(x, y), \quad v_x(x, y) = -u_y(x, y).$$

□

Example 1.11 Consider the function $f(z) = \bar{z}$.

For $z = x + iy$ we have $f(z) = x - iy$, i.e., $u(x, y) = x$, $v(x, y) = -y$. Thus, $u_x(x, y) = 1$ and $v_y(x, y) = -1$ for any $z = x + iy$. We see that the Cauchy-Riemann equations are not fulfilled. It follows that $f(z) = \bar{z}$ is not differentiable at any $z \in \mathbb{C}$.

△

Example 1.12 Consider the function $f(z) = |z|$.

We have already shown in [Example 1.7](#) that this function is not differentiable at $z = 0$. Using the Cauchy-Riemann equations, we will now show that $f(z) = |z|$ is not differentiable at any $z \in \mathbb{C}$.

We have $f(z) = |z| = \sqrt{x^2 + y^2}$, i.e. $u(x, y) = \sqrt{x^2 + y^2}$ and $v(x, y) = 0$. For $z \neq 0$ we have

$$\begin{aligned} u_x(x, y) &= \frac{x}{\sqrt{x^2 + y^2}}, & v_y(x, y) &= 0, \\ u_y(x, y) &= \frac{y}{\sqrt{x^2 + y^2}}, & v_x(x, y) &= 0, \end{aligned}$$

so that at least one of the Cauchy-Riemann equations does not hold at $z \neq 0$.

△

Remark Notice that the Cauchy-Riemann equations are necessary conditions for the differentiability of f at a given point. However, they are not sufficient. For example, the function

$$f(z) = \begin{cases} \frac{z^5}{|z|^4}, & z \neq 0, \\ 0, & z = 0, \end{cases}$$

satisfies the Cauchy-Riemann equations at $z = 0$, but is not differentiable at this point.

For those who are interested to see the details, we give them here. Using the binomial formula, we obtain

$$\begin{aligned} u(x, y) &= \begin{cases} \frac{x^5 - 10x^3y^2 + 5xy^4}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases} \\ v(x, y) &= \begin{cases} \frac{5x^4y - 10x^2y^3 + y^5}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases} \end{aligned}$$

Now observe that $u(x, 0) = x$, $u(0, y) = 0$, $v(x, 0) = 0$, $v(0, y) = y$ for all $(x, y) \in \mathbb{R}^2$. This implies that $u_x(0, 0) = v_y(0, 0) = 1$, $u_y(0, 0) = -v_x(0, 0) = 0$. Thus, the Cauchy-Riemann equations are fulfilled at $z = 0$.

However, f is not differentiable at $z = 0$. We consider the limit

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^4}{|h|^4}$$

which does not exist:

$$\lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{h^4}{|h|^4} = 1,$$

but for $h = k + ik$ with $k \rightarrow 0$, $k \in \mathbb{R}$ we have

$$\lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{(k + ik)^4}{|k + ik|^4} \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{-4k^4}{4k^4} = -1.$$

It is possible to formulate conditions that are necessary and sufficient, namely, we have to assume, in addition to the Cauchy-Riemann equations, that both functions u and v are differentiable as functions of two variables (this is more than just the existence of partial derivatives). We will not go into the detail.

△

1.3 The Mean Value Theorem

Recall the Mean Value Theorem of Real Analysis. It says that if $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

An important application is that if $f'(x) = 0$ for all x in an interval (a, b) , then f is constant on $[a, b]$.

The Mean Value Theorem does not hold for complex functions, as the next example shows.

Example 1.13 Consider the function $f(z) = z^4$, $z \in \mathbb{C}$. We have $f(1) = f(i) = 1$, so that

$$\frac{f(1) - f(i)}{1 - i} = 0.$$

However, $f'(z) = 4z^3$, and the only point with $f'(z) = 0$ is $z = 0$, which does not lie on the segment $[1, i]$.¹

△

¹We define a segment $[v, w]$ with $v, w \in \mathbb{C}$ by $[v, w] = \{z \in \mathbb{C} : z = \lambda v + (1 - \lambda)w, 0 \leq \lambda \leq 1\}$.

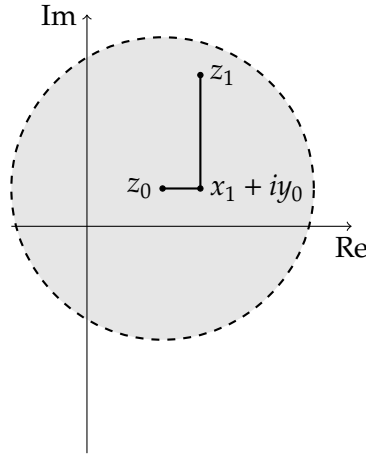


Figure 1.1: Proof of **Theorem 1.14**

Nevertheless, it is possible to prove that if $f'(z) = 0$ for all z in a proper set, then f is constant.

Theorem 1.14 Assume that f is holomorphic in an open disc $D(z_0; R)$ with some $z_0 \in \mathbb{C}$ and $R > 0$. If $f'(z) = 0$ for all $z \in D(z_0; R)$, then f is constant in $D(z_0; R)$.

△

Proof. We write $z = x + iy$ and $f = u + iv$. From the proof of **Theorem 1.10** we know that

$$f' = u_x + iv_x = -iu_y + v_y = 0.$$

It follows that $u_x = u_y = v_x = v_y = 0$ everywhere in $D(z_0; R)$.

Let $z_1 = x_1 + iy_1 \in D(z_0; R)$. We will show that $f(z_1) = f(z_0)$. We write $z_0 = x_0 + iy_0$ and consider the point $x_1 + iy_0$ that also lies in $D(z_0; R)$ (see **Figure 1.1**). The functions $x \mapsto u(x, y_0)$ and $x \mapsto v(x, y_0)$ are real-valued functions of a real variable whose derivatives vanish. By the Mean Value Theorem, they are constant on $[x_0, x_1]$. It follows that

$$u(x_0, y_0) = u(x_1, y_0), \quad v(x_0, y_0) = v(x_1, y_0).$$

Consequently, $f(z_0) = f(x_1 + iy_0)$.

Similarly, the functions $y \mapsto u(x_1, y)$ and $y \mapsto v(x_1, y)$ are real-valued function of a real variable whose derivatives vanish. By the Mean Value Theorem, they are constant on $[y_0, y_1]$. It follows that

$$u(x_1, y_0) = u(x_1, y_1), \quad v(x_1, y_0) = v(x_1, y_1).$$

It follows that $f(x_1 + iy_0) = f(z_1)$. Altogether we obtain $f(z_0) = f(z_1)$ as desired. $\square$

Clearly, the same proof will work for any open set with the property that any two points of the set are connected by paths made out of horizontal and vertical line segments.

1.4 On functional sequences and series

Before turning our attention to power series, we give the following definitions. They repeat verbatim the corresponding definitions for real functions.

Definition 1.15 Let $S \subset \mathbb{C}$ be a nonempty set. We say that a sequence of functions $f_n : S \rightarrow \mathbb{C}$, $n \in \mathbb{N}$, *converges pointwise* to $f : S \rightarrow \mathbb{C}$, if

$$\lim_{n \rightarrow \infty} f_n(z) = f(z) \quad \text{for all } z \in S.$$

$\triangle$

Definition 1.16 Let $S \subset \mathbb{C}$ be a nonempty set. We say that a sequence of functions $f_n : S \rightarrow \mathbb{C}$, $n \in \mathbb{N}$, *converges uniformly on S* to $f : S \rightarrow \mathbb{C}$, if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall z \in S \forall n \geq N : |f_n(z) - f(z)| < \varepsilon.$$

$\triangle$

Like in the real case, uniform convergence implies pointwise convergence, but not vice versa.

Definition 1.17 Let $(a_k)_{k=1}^{\infty}$ be a sequence of complex numbers. We say that the *series* $\sum_{k=1}^{\infty} a_k$ *converges to the sum* s if the sequence of its partial sums $(s_n)_{n=1}^{\infty}$, $s_n = \sum_{k=1}^n a_k$, converges to s .

$\triangle$

A straightforward consequence of the definition is the following statement.

Proposition 1.18 If $\sum_{k=1}^{\infty} a_k$ converges, then $a_k \rightarrow 0$ as $k \rightarrow \infty$.

$\triangle$

Example 1.19 A very important example is the series $\sum_{k=0}^{\infty} z^k$. If $|z| \geq 1$, then the terms of the series do not tend to zero, and therefore the series diverges. For $|z| < 1$ we have

$$s_n = \sum_{k=0}^n z^k = \frac{1 - z^{n+1}}{1 - z} \rightarrow \frac{1}{1 - z}, \quad n \rightarrow \infty.$$

In other words, for $|z| < 1$ the series converges and $\sum_{k=0}^{\infty} z^k = \frac{1}{1-z}$.

△

Definition 1.20 Let $(a_k)_{k=1}^{\infty}$ be a sequence of complex numbers. We say that the series $\sum_{k=1}^{\infty} a_k$ *converges absolutely* if the series $\sum_{k=1}^{\infty} |a_k|$ converges.

△

If a series converges absolutely, then it converges. There are several ways to prove this relatively simple statement. For example, one can consider $\operatorname{Re} a_n$ and $\operatorname{Im} a_n$ and apply known statements in the real case. Like in the real case, a convergent series need not converge absolutely.

Notice that $\sum_{k=1}^{\infty} |a_k|$ is a series of real numbers. This means that we can apply tests for convergence from Real Analysis (such as the comparison test, the ratio test, the root test and so on).

Our next statement is the Weierstrass M-Test that is, again, a repetition of the corresponding statement from Real Analysis.

Theorem 1.21 (Weierstrass M-Test) Let $S \subseteq \mathbb{C}$ be a nonempty set and $f_n : S \rightarrow \mathbb{C}$, $n \in \mathbb{N}$, be a sequence of complex functions. Suppose that for each $n \in \mathbb{N}$ there is a number $M_n \geq 0$ such that

$$|f_n(z)| \leq M_n \quad \text{for all } z \in S.$$

If $\sum_{n=1}^{\infty} M_n < \infty$, then the series $\sum_{n=1}^{\infty} f_n(z)$ converges absolutely and uniformly on S .

△

Proof. By the Comparison Test, the series $\sum_{n=1}^{\infty} |f_n(z)|$ converges for each $z \in S$. Thus, $\sum_{n=1}^{\infty} f_n(z)$ converges absolutely for every $z \in S$. It remains to prove that the convergence is uniform.

Put $f(z) = \sum_{n=1}^{\infty} f_n(z)$, $z \in S$; this function is well-defined on S . By the fact that the series $\sum_{n=1}^{\infty} M_n$ converges, given $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $\sum_{n=m+1}^{\infty} M_n < \varepsilon$ for all $m \geq N$. But then for any $m \geq N$ and any $z \in S$ we have

$$\left| f(z) - \sum_{n=1}^m f_n(z) \right| = \left| \sum_{n=m+1}^{\infty} f_n(z) \right| \leq \sum_{n=m+1}^{\infty} |f_n(z)| \leq \sum_{n=m+1}^{\infty} M_n < \varepsilon.$$

This proves the uniform convergence.

□

1.5 Power series

A series of the form $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ with $z_0 \in \mathbb{C}$, $(a_n)_{n=0}^{\infty} \subset \mathbb{C}$, is called a *power series*.

Definition 1.22 The *radius of convergence* of a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is the number

$$R = \sup \left\{ |z - z_0| : \sum_{n=0}^{\infty} a_n(z - z_0)^n \text{ converges} \right\}.$$

If $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges for all $z \in \mathbb{C}$, we put $R = \infty$.

△

The term “radius of convergence” will be justified by the theorem below.

Notice that every power series converges at $z = z_0$. Indeed,

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n \Big|_{z=z_0} = a_0 + a_1(z_0 - z_0) + a_2(z_0 - z_0)^2 + \cdots = a_0.$$

The case when the power series converges only for $z = z_0$ corresponds to the radius of convergence $R = 0$.

We will be mainly interested in power series with $R > 0$. Our first aim is to obtain a formula for R . First we need the following definition.

Definition 1.23 Let $(t_n)_{n=1}^{\infty}$ be a sequence of real numbers. Its *limit superior* (or *upper limit*) is defined by

$$\limsup_{n \rightarrow \infty} t_n = \lim_{n \rightarrow \infty} \left(\sup_{m \geq n} t_m \right).$$

For unbounded sequences, we allow values $\limsup_{n \rightarrow \infty} t_n = \pm\infty$.

△

If $(t_n)_{n=1}^{\infty}$ is bounded, its limit superior is exactly the supremum of limits of convergent subsequences of $(t_n)_{n=1}^{\infty}$. If $(t_n)_{n=1}^{\infty}$ converges, then its limit superior coincides with its limit. With our agreement to allow values $\pm\infty$, $\limsup_{n \rightarrow \infty} t_n$ always exists.

Theorem 1.24 (The Cauchy-Hadamard Theorem) Let $(a_n)_{n=0}^{\infty}$ be a sequence of complex numbers and $R > 0$ are such that

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} < \infty, \tag{1.1}$$

where we put $R = \infty$ if $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = 0$. The following holds true.

- (1) If $|z - z_0| < R$, then the power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges absolutely.
- (2) If $|z - z_0| > R$, then this series diverges.
- (3) The power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges absolutely and uniformly on any disc of the form $\{z \in \mathbb{C} : |z - z_0| \leq r\}$, where $0 \leq r < R$.

△

Equation (1.1) is known as the *Cauchy-Hadamard formula*. It says that the power series converges absolutely in the open disc $\{z \in \mathbb{C} : |z - z_0| < R\}$ and diverges outside of the closed disc $\{z \in \mathbb{C} : |z - z_0| \leq R\}$. At a point on the circle $\{z \in \mathbb{C} : |z - z_0| = R\}$, the power series might converge or diverge.

Proof of Theorem 1.24. Choose two real numbers r and s with $0 < r < s < R$. Then

$$\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \frac{1}{R} < \frac{1}{s}.$$

By the definition of the limit superior, there exists $N \in \mathbb{N}$ such that $|a_n|^{\frac{1}{n}} < \frac{1}{s}$ for all $n \geq N$, i.e.

$$|a_n| < \frac{1}{s^n} \quad \text{for all } n \geq N.$$

Let $|z - z_0| \leq r$. Then

$$|a_n(z - z_0)^n| \leq \left(\frac{r}{s}\right)^n \quad \text{for all } n \geq N.$$

Since the geometric series $\sum_{n=0}^{\infty} \left(\frac{r}{s}\right)^n$ converges, by the Weierstrass M-Test the series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges absolutely and uniformly in the disc $\{z \in \mathbb{C} : |z - z_0| \leq r\}$. This proves the first and the third statements of the theorem.

To prove the second statement, assume that $|z - z_0| > R$. Take two real numbers r and s with $R < s < r < |z - z_0|$. Now we have

$$\frac{1}{s} < \frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}.$$

By the definition of the limit superior, for any $n \in \mathbb{N}$ there is $m \geq n$ with $\frac{1}{s} < |a_m|^{\frac{1}{m}}$, i.e. $|a_m| > \frac{1}{s^m}$. But then

$$|a_m(z - z_0)^m| \geq \left(\frac{r}{s}\right)^m.$$

This inequality shows that the terms of the series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ do not converge to zero. It follows that the series diverges. $\square$

Example 1.25 Consider the power series $\sum_{n=0}^{\infty} \frac{n+1}{2n+1} (z+2)^n$.

We have

$$|a_n|^{\frac{1}{n}} = \left(\frac{n+1}{2n+1} \right)^{\frac{1}{n}} = e^{\frac{1}{n} \ln \left(\frac{n+1}{2n+1} \right)} \rightarrow e^0 = 1, \quad n \rightarrow \infty.$$

Thus, in this case we even have the limit $\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = 1$. By the Cauchy-Hadamard formula, $R = 1$. $\triangle$

Example 1.26 Consider the power series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} z^{2n+1}$; this is the Taylor series for the function $\arctan z$.

Here we have

$$a_m = \begin{cases} \frac{(-1)^n}{2n+1}, & m = 2n+1 \text{ is odd,} \\ 0, & m = 2n \text{ is even.} \end{cases}$$

Clearly, $\lim_{n \rightarrow \infty} |a_{2n}|^{\frac{1}{2n}} = 0$. For the subsequence of the terms with odd indices we have

$$|a_{2n+1}|^{\frac{1}{2n+1}} = \left(\frac{1}{2n+1} \right)^{\frac{1}{2n+1}} = e^{\frac{1}{2n+1} \ln \left(\frac{1}{2n+1} \right)} \rightarrow e^0 = 1, \quad n \rightarrow \infty.$$

We conclude that $\limsup_{m \rightarrow \infty} |a_m|^{\frac{1}{m}} = 1$ and $R = 1$.

Notice that the limit $\lim_{m \rightarrow \infty} |a_m|^{\frac{1}{m}}$ does not exist. $\triangle$

The next result is based on the Ratio Test from Real Analysis and is in some cases useful for calculating the radius of convergence of a power series.

Proposition 1.27 (The Ratio Formula) If $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} < \infty$, then the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ can be determined by the formula

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}.$$

$\triangle$

The proof repeats the proof of an analogous statement from Real Analysis and is left as an exercise.

Example 1.28 Consider the power series $\sum_{n=0}^{\infty} \frac{z^n}{n!}$.

We have

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n \rightarrow \infty} \frac{n!}{(n+1)!} = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0,$$

so that $R = \infty$.

△

If a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has a non-zero radius of convergence R , then it converges in the open disc $D(z_0; R) = \{z \in \mathbb{C} : |z - z_0| < R\}$ and defines in $D(z_0; R)$ a function $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$. We are now approaching a very important result that this function f is holomorphic in $D(z_0; R)$.

Lemma 1.29 Let $(\alpha_n)_{n=0}^{\infty}$ and $(\beta_n)_{n=0}^{\infty}$ be two sequences in $[0, \infty)$ such that

$$\lim_{n \rightarrow \infty} \alpha_n = \alpha \in (0, \infty), \quad \limsup_{n \rightarrow \infty} \beta_n = \beta \in [0, \infty).$$

Then

$$\limsup_{n \rightarrow \infty} \alpha_n \beta_n = \alpha \beta.$$

△

Proof. If $\limsup_{n \rightarrow \infty} \beta_n = 0$, this together with the fact that $\beta_n \geq 0$ for all $n \in \mathbb{N}$ implies that $\lim_{n \rightarrow \infty} \beta_n = 0$. Then the statement follows from the known rules for limits of real sequences.

Now assume that $\beta > 0$. Given $\lambda > 1$, there is $N \in \mathbb{N}$ such that

$$0 < \frac{1}{\lambda} \alpha < \alpha_n < \lambda \alpha \quad \text{and} \quad \beta_n < \lambda \beta \quad \text{for all } n \geq N.$$

Moreover, for any $m \in \mathbb{N}$ there is $n \geq m$ with

$$\frac{1}{\lambda} \beta < \beta_n.$$

(Please make sure that you understand this.) It follows that

$$\frac{1}{\lambda^2} \alpha \beta < \sup_{n \geq N} \alpha_n \beta_n < \lambda^2 \alpha \beta.$$

This gives $\limsup_{n \rightarrow \infty} \alpha_n \beta_n = \alpha \beta$.

□

Theorem 1.30 *If a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has radius of convergence $R > 0$, then the term-by-term differentiated power series $\sum_{n=1}^{\infty} a_n n(z - z_0)^{n-1}$ has the same radius of convergence R .*

△

Proof. Notice that for a fixed $z \in \mathbb{C}$

$$(z - z_0) \sum_{n=1}^{\infty} a_n (z - z_0)^{n-1} = \sum_{n=0}^{\infty} a_n (z - z_0)^n - a_0,$$

and thus the series $\sum_{n=1}^{\infty} a_n (z - z_0)^{n-1}$ and $\sum_{n=0}^{\infty} a_n (z - z_0)^n - a_0$ either both converge or both diverge at z . This implies that they have the same radius of convergence R .

By the Cauchy-Hadamard Theorem ([Theorem 1.24](#)),

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}.$$

On the other hand, the radius of convergence of $\sum_{n=1}^{\infty} a_n n(z - z_0)^{n-1}$ is

$$\frac{1}{R_1} = \limsup_{n \rightarrow \infty} |a_n n|^{\frac{1}{n-1}}.$$

Since $\lim_{n \rightarrow \infty} n^{\frac{1}{n-1}} = 1$, by [Lemma 1.29](#) we obtain that $\frac{1}{R_1} = \frac{1}{R}$, as desired. □

Theorem 1.31 *Let $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ be a power series with radius of convergence $R > 0$. Then the function $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ is holomorphic in $D(z_0; R)$, and*

$$f'(z) = \sum_{n=1}^{\infty} a_n n(z - z_0)^{n-1}, \quad z \in D(z_0; R).$$

△

Proof. By [Theorem 1.30](#), the function $g(z) = \sum_{n=1}^{\infty} a_n n(z - z_0)^{n-1}$ is well-defined on $D(z_0; R)$. We have to prove that $f'(z)$ exists, and $f'(z) = g(z)$.

Without loss of generality, assume that $z_0 = 0$. Let $z, z + h \in D(0; R)$. Using the binomial formula

$$(z + h)^n = z^n + \sum_{k=1}^n \binom{n}{k} z^{n-k} h^k,$$

we estimate

$$\begin{aligned}
 \left| \frac{f(z+h) - f(z)}{h} - g(z) \right| &= \left| \sum_{n=0}^{\infty} a_n \frac{(z+h)^n - z^n}{h} - \sum_{n=1}^{\infty} a_n n z^{n-1} \right| \\
 &= \left| \sum_{n=1}^{\infty} a_n \left(\frac{(z+h)^n - z^n}{h} - n z^{n-1} \right) \right| \\
 &= \left| \sum_{n=1}^{\infty} a_n \left(\sum_{k=1}^n \binom{n}{k} z^{n-k} h^{k-1} - n z^{n-1} \right) \right| \\
 &= \left| \sum_{n=1}^{\infty} a_n \sum_{k=2}^n \binom{n}{k} z^{n-k} h^{k-1} \right| \\
 &\leq |h| \sum_{n=1}^{\infty} |a_n| \sum_{k=2}^n \binom{n}{k} |z|^{n-k} |h|^{k-2} \\
 &= |h| \sum_{n=1}^{\infty} |a_n| \sum_{m=0}^{n-2} \binom{n}{m+2} |z|^{n-2-m} |h|^m.
 \end{aligned}$$

For the binomial coefficient we have

$$\begin{aligned}
 \binom{n}{m+2} &= \frac{n!}{(m+2)!(n-m-2)!} = \frac{n(n-1)(n-2)!}{(m+2)(m+1)m!(n-2-m)!} \\
 &= \frac{n(n-1)}{(m+2)(m+1)} \binom{n-2}{m} \leq \frac{n(n-1)}{2} \binom{n-2}{m}.
 \end{aligned}$$

We substitute this into the above estimate and obtain

$$\begin{aligned}
 \left| \frac{f(z+h) - f(z)}{h} - g(z) \right| &\leq |h| \sum_{n=1}^{\infty} |a_n| \frac{n(n-1)}{2} \sum_{m=0}^{n-2} \binom{n-2}{m} |z|^{n-2-m} |h|^m \\
 &= \frac{|h|}{2} \sum_{n=1}^{\infty} n(n-1) |a_n| (|z| + |h|)^{n-2}.
 \end{aligned}$$

Now we fix z with $|z| < R$. Take $r > 0$ with $|z| < r < R$. Applying [Theorem 1.30](#) two times, we see that the series $\sum_{n=1}^{\infty} n(n-1) |a_n| r^{n-2}$ converges. Put $M = \sum_{n=1}^{\infty} n(n-1) |a_n| r^{n-2}$. If $|h|$ is so small that $z+h \in D(0; R)$ and $|z| + |h| < r$, then

$$\left| \frac{f(z+h) - f(z)}{h} - g(z) \right| \leq \frac{1}{2} M |h|.$$

Thus,

$$\lim_{h \rightarrow 0} \left| \frac{f(z+h) - f(z)}{h} - g(z) \right| = 0.$$

This implies that

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = g(z).$$

□

Substituting $z = z_0$ into the formula $f'(z) = \sum_{n=1}^{\infty} a_n n (z - z_0)^{n-1}$, we see that $f'(z_0) = a_1$. Now we can apply **Theorem 1.31** to the power series $\sum_{n=1}^{\infty} a_n n (z - z_0)^{n-1}$ to conclude that f'' exists in $D(0; R)$ and to obtain a power series representation for f'' . Iterating the procedure, we arrive at the following statement.

Corollary 1.32 *Let $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ be a power series with radius of convergence $R > 0$. Then the function $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ has derivatives of all orders in $D(z_0; R)$, and*

$$f^{(n)}(z_0) = n! a_n, \quad n = 0, 1, 2, \dots$$

△

In particular, the coefficients in the power series expansion are unique, and the power series is exactly the Taylor series of f .

A function f on an open set G is called **analytic** on G , if for any $z_0 \in G$ f is represented by a power series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ that converges to $f(z)$ in some neighborhood of the point z_0 . **Theorem 1.31** says that an analytic function is holomorphic. We will see later that also the converse is true: a complex function is holomorphic if and only if it is analytic. Because of this fact, the terms “holomorphic” and “analytic” are frequently used interchangeably for complex functions. However, the term “analytic” can be applied more generally, i.e., not only to complex functions.

1.6 Elementary functions

Definition 1.33 The **exponential function** is defined by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad z \in \mathbb{C}.$$

△

We have seen in **Example 1.28** that the power series $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ converges for all $z \in \mathbb{C}$.

Theorem 1.34 (*Properties of the exponential function*)

(1) The function e^z is holomorphic (and hence continuous) on $\mathbb{C}$, and $(e^z)' = e^z$,

- (2) $e^{z+w} = e^z e^w$ for $z, w \in \mathbb{C}$,
- (3) $e^z \neq 0$ for any $z \in \mathbb{C}$; $e^x > 0$ for any $x \in \mathbb{R}$,
- (4) $|e^z| = e^{\operatorname{Re} z}$ for $z \in \mathbb{C}$,
- (5) $|e^{it}| = 1$ for $t \in \mathbb{R}$.

△

Proof. (1) follows immediately from **Theorem 1.31**.

- (2) First notice that $e^0 = 1$; this follows easily by substituting 0 in the power series in **Definition 1.33**.

For fixed $\zeta \in \mathbb{C}$, consider the function $g(z) = e^z e^{\zeta-z}$. For its derivative we obtain, using the chain rule,

$$g'(z) = e^z e^{\zeta-z} - e^z e^{\zeta-z} = 0.$$

Thus, by **Theorem 1.14** g has the constant value $g(0) = e^0 e^\zeta = e^\zeta$ on $\mathbb{C}$, i.e.

$$e^\zeta = e^z e^{\zeta-z}.$$

We take $\zeta = z + w$ and obtain $e^{z+w} = e^z e^w$.

- (3) For $z \in \mathbb{C}$ we have by (2) $e^z e^{-z} = e^0 = 1$. It follows that $e^z \neq 0$. If x is real and positive, then $e^x > 0$ by the form of the power series (all terms are positive). In this case we also have $e^{-x} = \frac{1}{e^x} > 0$.

- (4) Notice that

$$\overline{e^z} = \overline{\sum_{n=1}^{\infty} \frac{z^n}{n!}} = \sum_{n=1}^{\infty} \frac{\overline{z}^n}{n!} = e^{\overline{z}}.$$

Using this fact, we obtain

$$|e^z|^2 = e^z \overline{e^z} = e^z e^{\overline{z}} = e^{z+\overline{z}} = e^{2\operatorname{Re} z} = e^{\operatorname{Re} z + \operatorname{Re} z} = (e^{\operatorname{Re} z})^2.$$

Since both numbers $|e^z|$ and $e^{\operatorname{Re} z}$ are real and positive, this implies $|e^z| = e^{\operatorname{Re} z}$.

- (5) If $t \in \mathbb{R}$, then $\operatorname{Re}(it) = 0$, and thus $|e^{it}| = e^0 = 1$.

□

Definition 1.35 The trigonometric functions *cosine* and *sine* are defined by

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

△

It follows immediately from the definitions that

$$e^{iz} = \cos z + i \sin z, \quad z \in \mathbb{C}.$$

Substituting the power series for e^{iz} into the definitions of cosine and sine, we obtain their power series expansions

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \quad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$$

(Check this as an exercise.) Both series converge absolutely for any $z \in \mathbb{C}$. Differentiating these series term-by-term shows that

$$(\cos z)' = -\sin z, \quad (\sin z)' = \cos z.$$

For real z , the functions e^z , $\cos z$, $\sin z$ coincide with their real counterparts that are well-known to you. The addition formulas, etc., are also valid for complex sine and cosine functions. However, the inequalities $|\sin t| \leq 1$, $|\cos t| \leq 1$, $t \in \mathbb{R}$, do not hold anymore for complex sine and cosine functions. For example, if $t \in \mathbb{R}$, then

$$\cos it = \frac{1}{2}(e^t + e^{-t}) \rightarrow \infty \quad \text{as } t \rightarrow \infty,$$

and thus the function $\cos z$ is unbounded on $\mathbb{C}$. Similarly, the function $\sin z$ is unbounded on $\mathbb{C}$.

If $z = x + iy$ with $x = \operatorname{Re} z$, $y = \operatorname{Im} z$, then it follows from the addition formula for the exponential function that

$$e^z = e^x e^{iy} = e^x (\cos y + i \sin y).$$

In particular, for $\theta \in \mathbb{R}$ we recover the well-known formula

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

We have already mentioned that $e^z \neq 0$ for any $z \in \mathbb{C}$. Notice that

$$e^z = e^x (\cos y + i \sin y) = 1$$

Chapter 1 Holomorphic functions

if and only if $x = 0, y = 2\pi k, k \in \mathbb{Z}$, i.e.

$$e^z = 1 \iff z = 2\pi ki, k \in \mathbb{Z}.$$

It follows that

$$e^{z+w} = e^z \iff w = 2\pi ki, k \in \mathbb{Z}.$$

Similarly, one can show that

$$\cos z = 0 \iff z = \frac{\pi}{2} + \pi k, k \in \mathbb{Z},$$

and

$$\sin z = 0 \iff z = \pi k, k \in \mathbb{Z}.$$

Chapter 2

Integrals of complex functions

2.1 Integrals along paths

Let $[a, b] \subset \mathbb{R}$ be a compact interval. A *curve* γ with parameter interval $[a, b]$ is a continuous function $\gamma : [a, b] \rightarrow \mathbb{C}$. The point $\gamma(a)$ is called its *initial point*, and the point $\gamma(b)$ is called its *end point*. The curve γ is called *closed* if $\gamma(a) = \gamma(b)$. The curve is called *simple* if $\gamma(t) \neq \gamma(s)$ for all $t, s \in (a, b)$, $t \neq s$.

The image set $\{\gamma(t) : t \in [a, b]\}$ is a compact subset of $\mathbb{C}$ (as a continuous image of a compact interval $[a, b]$). Notice that the same image set can have different parametrizations. For example the unit circle in $\mathbb{C}$ can be parameterized as

$$\begin{aligned}\gamma_1(t) &= e^{it}, & t \in [0, 2\pi], \\ \gamma_2(t) &= e^{-it}, & t \in [0, 2\pi], \\ \gamma_3(t) &= e^{2\pi it}, & t \in [0, 1],\end{aligned}$$

and so on.

A curve $\gamma : [a, b] \rightarrow \mathbb{C}$ is called *smooth*, if γ has a continuous derivative on $[a, b]$. Here we consider γ as a complex-valued function of a real variable: $\gamma(t) = \operatorname{Re} \gamma(t) + i \operatorname{Im} \gamma(t)$, and its derivative is understood as $\gamma'(t) = (\operatorname{Re} \gamma)'(t) + i(\operatorname{Im} \gamma)'(t)$. The derivatives at the points a and b are the one-sided derivatives.

Suppose that we have two curves γ_1, γ_2 such that the end point of γ_1 coincides with the initial point of γ_2 . Without loss of generality, we assume that γ_1 and γ_2 have parameter intervals $[a_0, a_1]$ and $[a_1, a_2]$, respectively. Then we can build their *join*

$$\gamma(t) = \begin{cases} \gamma_1(t), & t \in [a_0, a_1], \\ \gamma_2(t), & t \in [a_1, a_2]. \end{cases}$$

The joining process can be iterated to build a join of finitely many curves $\gamma_1, \gamma_2, \dots, \gamma_n$, with the end point of γ_k being the initial point of γ_{k+1} , $k = 1, \dots, n-1$.

Definition 2.1 A *path of integration (contour)* is a join of finitely many smooth curves. Δ

Notice that a path of integration γ is piecewise continuously differentiable: if γ is the joint of smooth curves $\gamma_1, \dots, \gamma_n$ with parameter intervals $[a_{k-1}, a_k]$, $k = 1, \dots, n$, then $\gamma'(t) = \gamma'_k(t)$ for $t \in (a_{k-1}, a_k)$. The derivative $\gamma'(t)$ may not exist at the points a_k , $k = 0, \dots, n$.

Let $S \subseteq \mathbb{C}$. We say that γ is a *path in S* , if $\{\gamma(t) : t \in [a, b]\} \subseteq S$.

If $g : [a, b] \rightarrow \mathbb{C}$ is $g = \operatorname{Re} g + i \operatorname{Im} g$ and the real-valued functions $\operatorname{Re} g$, $\operatorname{Im} g$ are integrable, then we put

$$\int_a^b g(t) dt = \int_a^b \operatorname{Re} g(t) dt + i \int_a^b \operatorname{Im} g(t) dt.$$

Notice that a piecewise continuous function is integrable: if g coincides with a continuous function g_k on each interval $[a_{k-1}, a_k]$, $k = 1, \dots, n$, then

$$\int_{a_0}^{a_n} g(t) dt = \sum_{k=1}^n \int_{a_{k-1}}^{a_k} g(t) dt.$$

The values of g at the points a_k , $k = 0, \dots, n$, need not be defined.

Definition 2.2 Let $f : S \rightarrow \mathbb{C}$ be a continuous function on $S \subseteq \mathbb{C}$, and let γ be a path in S with parameter interval $[a, b]$. The *integral of f along the path γ* is defined by

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Δ

Remark For the definition of the integral, it is sufficient to assume that f is a continuous function defined on the set $\{\gamma(t) : t \in [a, b]\}$.

Δ

Example 2.3 Let $a \in \mathbb{C}$ and $r > 0$. The curve $\gamma(t) = a + re^{it}$, $t \in [0, 2\pi]$, is a circle with center a and radius r . Take the function $f(z) = (z - a)^n$, $n \in \mathbb{Z}$. We have

$$\int_{\gamma} (z - a)^n dz = \int_0^{2\pi} (re^{it})^n rie^{it} dt = ir^{n+1} \int_0^{2\pi} e^{i(n+1)t} dt.$$

Now if $n = -1$, then $e^{i(n+1)t} \equiv 1$, and $\int_0^{2\pi} e^{i(n+1)t} dt = 2\pi$. If $n \neq -1$, we have

$$\int_0^{2\pi} e^{i(n+1)t} dt = \int_0^{2\pi} \cos((n+1)t) dt + i \int_0^{2\pi} \sin((n+1)t) dt = 0.$$

Thus,

$$\int_{\gamma} (z - a)^n dz = \begin{cases} 0, & n \neq -1, \\ 2\pi i, & n = -1. \end{cases}$$

△

The next two simple properties of the integral follow immediately from the definition of the integral and the properties of the integral of real-valued functions. For a path $\gamma(t)$, $t \in [a, b]$, we denote by $-\gamma$ the path

$$(-\gamma)(t) = \gamma(a + b - t), \quad t \in [a, b];$$

this path has the same image set as γ , but the opposite orientation.

Proposition 2.4 *Let $f : S \rightarrow \mathbb{C}$ be a continuous function and γ be a path in S .*

(1) *If γ is a join of two paths γ_1 and γ_2 , then*

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz.$$

(2)

$$\int_{-\gamma} f(z) dz = - \int_{\gamma} f(z) dz.$$

△

We have seen that the same set of points in $\mathbb{C}$ may have different parameterizations. In the next statement we consider an effect of reparameterization.

Lemma 2.5 *Let γ be a path in S with parameter interval $[a, b]$. Let $\psi : [\tilde{a}, \tilde{b}] \rightarrow [a, b]$ be a function with $\psi(\tilde{a}) = a$, $\psi(\tilde{b}) = b$ and a continuous derivative $\psi' > 0$. Then for any continuous $f : S \rightarrow \mathbb{C}$ we have*

$$\int_{\gamma \circ \psi} f(z) dz = \int_{\gamma} f(z) dz.$$

△

Notice that $\gamma \circ \psi$ is a path with parameter interval $[\tilde{a}, \tilde{b}]$, and that the image sets of γ and of $\gamma \circ \psi$ is the same. The assumption $\psi' > 0$ guarantees that $\gamma \circ \psi$ has the same orientation as γ .

Proof. By the chain rule, $(\gamma \circ \psi)'(t) = \gamma'(\psi(t))\psi'(t)$ for $t \in [\tilde{a}, \tilde{b}]$. Making the substitution $s = \psi(t)$, we have

$$\begin{aligned} \int_{\gamma \circ \psi} f(z) dz &= \int_{\tilde{a}}^{\tilde{b}} f((\gamma \circ \psi)(t))(\gamma \circ \psi)'(t) dt = \int_{\tilde{a}}^{\tilde{b}} f(\gamma(\psi(t)))\gamma'(\psi(t))\psi'(t) dt \\ &= \int_a^b f(\gamma(s))\gamma'(s) ds = \int_{\gamma} f(z) dz. \end{aligned}$$

□

Thus, under mild conditions the integral depends only on the image set $\{\gamma(t) : t \in [a, b]\}$ and on the direction in which it is traced, but not on the concrete choice of parameterization.

Theorem 2.6 (The Estimation Theorem) *Let γ be a path in S with parameter interval $[a, b]$, and let $f : S \rightarrow \mathbb{C}$ be a continuous function. Then*

$$\left| \int_{\gamma} f(z) dz \right| \leq \int_a^b |f(\gamma(t))\gamma'(t)| dt.$$

△

Proof. It is enough to consider the case when γ is a smooth curve. Since $\int_{\gamma} f(z) dz$ is a complex number, it can be written in the form

$$\int_{\gamma} f(z) dz = re^{i\theta}, \quad \text{where } r = \left| \int_{\gamma} f(z) dz \right| \text{ and } \theta \in \mathbb{R}.$$

Then we have

$$\left| \int_{\gamma} f(z) dz \right| = r = e^{-i\theta} \int_{\gamma} f(z) dz = \int_{\gamma} e^{-i\theta} f(z) dz.$$

Notice that this is a non-negative real number. Therefore,

$$\begin{aligned} \left| \int_{\gamma} f(z) dz \right| &= \operatorname{Re} \left(\int_{\gamma} e^{-i\theta} f(z) dz \right) = \operatorname{Re} \left(\int_a^b e^{-i\theta} f(\gamma(t))\gamma'(t) dt \right) \\ &= \int_a^b \operatorname{Re} (e^{-i\theta} f(\gamma(t))\gamma'(t)) dt \leq \int_a^b |e^{-i\theta} f(\gamma(t))\gamma'(t)| dt \\ &= \int_a^b |f(\gamma(t))\gamma'(t)| dt, \end{aligned}$$

in the last step we used the fact that $|e^{-i\theta}| = 1$.

□

Corollary 2.7 Let γ be a path in S with parameter interval $[a, b]$, and let $f : S \rightarrow \mathbb{C}$ be a continuous function satisfying $|f(z)| \leq M$ for all z in the image set of γ . Then

$$\left| \int_{\gamma} f(z) dz \right| \leq M \int_a^b |\gamma'(t)| dt.$$

△

This statement follows from **Theorem 2.6** in an obvious way. For a path γ , we define the value $\ell(\gamma) = \int_a^b |\gamma'(t)| dt$ to be the *length* of γ .

Example 2.8 Consider $f(z) = \frac{1}{z}$ and $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. By **Example 2.3**,

$$\int_{\gamma} f(z) dz = 2\pi i.$$

We have $|f(\gamma(t))| = 1$ and $|\gamma'(t)| = |ie^{it}| = 1$. The Estimation Theorem (**Theorem 2.6**) gives

$$\left| \int_{\gamma} f(z) dz \right| \leq \int_0^{2\pi} 1 dt = 2\pi.$$

Notice that the integral $\int_{\gamma} |f(z)| dz$ does not give an upper estimate for $\left| \int_{\gamma} f(z) dz \right|$. Indeed,

$$\int_{\gamma} |f(z)| dz = \int_{\gamma} 1 dz = \int_0^{2\pi} \gamma'(t) dt = \int_0^{2\pi} ie^{it} dt = \int_0^{2\pi} (i \cos t - \sin t) dt = 0.$$

△

2.2 The fundamental theorem of calculus

Theorem 2.9 (The Fundamental Theorem of Calculus) Let $G \subseteq \mathbb{C}$ be an open set, let γ be a path in G , and let $f : G \rightarrow \mathbb{C}$ be a continuous function in G . If there exists a function $F : G \rightarrow \mathbb{C}$ such that $F'(z) = f(z)$, $z \in G$, then

$$\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a)).$$

In particular, if γ is closed, then

$$\int_{\gamma} f(z) dz = 0.$$

△

Proof. First we assume that γ is smooth. Notice that the function $F \circ \gamma$ is differentiable on $[a, b]$, with $(F \circ \gamma)'(t) = F'(\gamma(t))\gamma'(t)$. Applying the real Fundamental Theorem of Calculus, we obtain

$$\begin{aligned} \int_{\gamma} f(z) dz &= \int_{\gamma} F'(z) dz = \int_a^b F'(\gamma(t))\gamma'(t) dt = \int_a^b (F \circ \gamma)'(t) dt \\ &= \int_a^b (\operatorname{Re}(F \circ \gamma))'(t) dt + i \int_a^b (\operatorname{Im}(F \circ \gamma))'(t) dt \\ &= \operatorname{Re}(F(\gamma(b))) - \operatorname{Re}(F(\gamma(a))) + i(\operatorname{Im}(F(\gamma(b))) - \operatorname{Im}(F(\gamma(a)))) \\ &= F(\gamma(b)) - F(\gamma(a)). \end{aligned}$$

If γ is a piecewise smooth path, take $a = t_0 < t_1 < \dots < t_n = b$ such that the restriction of γ to each interval $[t_{k-1}, t_k]$, $k = 1, \dots, n$, is smooth. Applying the above formula to each interval, we obtain

$$\begin{aligned} \int_{\gamma} f(z) dz &= \sum_{k=1}^n \int_{t_{k-1}}^{t_k} F'(\gamma(t))\gamma'(t) dt \\ &= \sum_{k=1}^n (F(\gamma(t_k)) - F(\gamma(t_{k-1}))) = F(\gamma(b)) - F(\gamma(a)). \end{aligned}$$

□

In contrast to the real case, the complex Fundamental Theorem of Calculus can be considered an interim result. There is a much more powerful statement — Cauchy's Integral Theorem. We will formulate and prove one version of it in the next chapter.

Example 2.10 Consider the function $f(z) = z$. It has an antiderivative $F(z) = \frac{1}{2}z^2$ everywhere on $\mathbb{C}$. Thus, if γ is any closed path in $\mathbb{C}$, then $\int_{\gamma} z dz = 0$.

Compare this situation with the function $g(z) = \bar{z}$. Let $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. We have

$$\int_{\gamma} \bar{z} dz = \int_0^{2\pi} e^{-it} i e^{it} dt = 2\pi i.$$

The integral is not zero; we conclude that $g(z) = \bar{z}$ does not have an antiderivative. We will see later that this behavior is connected to the fact that $g(z) = \bar{z}$ is not holomorphic.

△

Chapter 3

Cauchy's integral theorem

Cauchy's Integral Theorem says, roughly speaking, that if G is an open set in $\mathbb{C}$, γ is a closed path in G and f is a function holomorphic in G , then $\int_{\gamma} f(z) dz = 0$. There exist different variants of this theorem; the difference is in the exact assumptions on the open set G and on the path γ . We will consider here one variant.

We have seen in the previous chapter that $\int_{\gamma} f(z) dz = 0$ if f has an antiderivative in G . So it will not be a surprise that we will work with antiderivatives.

We start with a result for a triangle.

Theorem 3.1 (Goursat's Lemma) *Let f be holomorphic on an open set $G \subseteq \mathbb{C}$, and let Δ be a closed triangle in G . Let γ be the path along the edges of Δ . Then*

$$\int_{\gamma} f(z) dz = 0.$$

△

Proof. Put $I = \int_{\gamma} f(z) dz$.

Put $\Delta_0 = \Delta$, $\gamma_0 = \gamma$. We subdivide Δ_0 into four triangles $\Delta_1^1, \dots, \Delta_1^4$ by connecting the midpoints of the edges of Δ_0 . We denote by $\gamma_1^1, \dots, \gamma_1^4$ the paths along the edges of $\Delta_1^1, \dots, \Delta_1^4$, respectively. Then by **Proposition 2.4** we have

$$I = \int_{\gamma_0} f(z) dz = \sum_{k=1}^4 \int_{\gamma_1^k} f(z) dz.$$

For at least one $k = 1, \dots, 4$ we will have $\left| \int_{\gamma_1^k} f(z) dz \right| \geq \frac{1}{4} |I|$. We take this Δ_1^k to be Δ_1 and denote γ_1^k by γ_1 . Then we apply the same procedure to Δ_1 to construct Δ_2 and γ_2 . Continuing in this manner, we obtain a sequence of nested triangles

$$\Delta_0 \supseteq \Delta_1 \supseteq \Delta_2 \supseteq \Delta_3 \supseteq \dots$$

with the property that

$$|I| \leq 4^n \left| \int_{\gamma_n} f(z) dz \right|, \quad n = 0, 1, \dots$$

We denote by $\ell(\gamma_n)$ the length of the path γ_n . By construction, $\ell(\gamma_n) = 2^{-n}\ell(\gamma)$, $n = 0, 1, \dots$

Notice that all Δ_n are compact. Therefore, $\bigcap_{n=0}^{\infty} \Delta_n \neq \emptyset$. Take a point $z_0 \in \bigcap_{n=0}^{\infty} \Delta_n$.

The function f is differentiable at z_0 . By **Proposition 1.5**, there exists a function $\epsilon(z)$ such that

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \epsilon(z)(z - z_0), \quad z \in G,$$

where $\lim_{z \rightarrow z_0} \epsilon(z) = 0$.

Given $\varepsilon > 0$, there exists $\delta > 0$ such that $|\epsilon(z)| < \varepsilon$ for all $z \in D(z_0; \delta)$. Choose $N \in \mathbb{N}$ such that $\Delta_N \subseteq D(z_0; \delta)$. Then

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| < \varepsilon |z - z_0| \quad \text{for all } z \in \Delta_N.$$

Moreover, for $z \in \Delta_N$ we have $|z - z_0| \leq \ell(\gamma_N) = 2^{-N}\ell(\gamma)$, so that

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| < \varepsilon 2^{-N}\ell(\gamma) \quad \text{for all } z \in \Delta_N.$$

By **Corollary 2.7**,

$$\left| \int_{\gamma_N} (f(z) - f(z_0) - f'(z_0)(z - z_0)) dz \right| \leq \varepsilon 2^{-N}\ell(\gamma)\ell(\gamma_N) = \varepsilon (2^{-N}\ell(\gamma))^2.$$

The function $f(z_0) + f'(z_0)(z - z_0)$ is a linear function of z . By **Theorem 2.9**,

$$\int_{\gamma_N} (f(z_0) + f'(z_0)(z - z_0)) dz = 0.$$

Hence,

$$\left| \int_{\gamma_N} f(z) dz \right| \leq \varepsilon (2^{-N}\ell(\gamma))^2 = \varepsilon 4^{-N}(\ell(\gamma))^2.$$

This implies

$$|I| \leq 4^N \left| \int_{\gamma_N} f(z) dz \right| \leq \varepsilon (\ell(\gamma))^2.$$

Since this is true for any $\varepsilon > 0$, we have $I = 0$.

□

Now we are going to specify the conditions on G .

Definition 3.2 An open set $G \subseteq \mathbb{C}$ is called *connected*, if G cannot be represented as $G_1 \cup G_2$, where G_1 and G_2 are two non-empty open sets with $G_1 \cap G_2 = \emptyset$. $\triangle$

Definition 3.3 A *region* is a non-empty open connected subset of $\mathbb{C}$. $\triangle$

Definition 3.4 A region $G \subseteq \mathbb{C}$ is called *star-shaped*, if there exists the point $a \in G$ such that for any $z \in G$ the segment $[a, z]$ is contained in G . $\triangle$

A point $a \in G$ with this property is in general not unique. Notice that every convex region G is star-shaped; in this case any point of G can be taken for a with the above property.

Lemma 3.5 Let f be a continuous function on a star-shaped region $G \subseteq \mathbb{C}$ such that

$$\int_{\gamma} f(z) dz = 0$$

for any triangle γ in G . Let a be such that $[a, z] \subseteq G$ for any $z \in G$. Then the function $F : G \rightarrow \mathbb{C}$,

$$F(z) = \int_{[a,z]} f(w) dw$$

is holomorphic in G , with $F' = f$. $\triangle$

Proof. Fix $z \in G$. Take $r > 0$ so small that $D(z; r) \subseteq G$. Then $|h| < r$ implies that $[z, z+h] \subseteq G$. Since G is star-shaped, both segments $[a, z]$ and $[a, z+h]$ lie in G . Therefore, the triangle with the vertices a, z and $z+h$ lies in G . Let γ be a path along the edges of this triangle. By assumption,

$$\int_{\gamma} f(w) dw = \int_{[a,z]} f(w) dw + \int_{[z,z+h]} f(w) dw - \int_{[a,z+h]} f(w) dw = 0.$$

It follows that

$$F(z+h) - F(z) = \int_{[a,z+h]} f(w) dw - \int_{[a,z]} f(w) dw = \int_{[z,z+h]} f(w) dw.$$

Notice that the segment $[z, z+h]$ can be parameterized as $z+ht$, $t \in [0, 1]$. In particular, for the constant function $f(z)$

$$\int_{[z,z+h]} f(z) dw = \int_0^1 f(z) h dt = f(z)h.$$

We have

$$\begin{aligned}
 \left| \frac{F(z+h) - F(z)}{h} - f(z) \right| &= \left| \frac{1}{h} \int_{[z, z+h]} f(w) dw - \frac{1}{h} \int_{[z, z+h]} f(z) dw \right| \\
 &= \left| \frac{1}{h} \int_{[z, z+h]} (f(w) - f(z)) dw \right| \\
 &\leq \frac{1}{|h|} \int_0^1 |f(z+ht) - f(z)| |h| dt \\
 &= \int_0^1 |f(z+ht) - f(z)| dt \\
 &\leq \sup_{t \in [0,1]} |f(z+ht) - f(z)| \rightarrow 0, \quad h \rightarrow 0,
 \end{aligned}$$

by the continuity of f at z . Thus, F is differentiable at z with $F'(z) = f(z)$. □

Theorem 3.6 (*Cauchy's Integral Theorem for star-shaped regions*) *Let G be a star-shaped region, and let f be a holomorphic function on G . Then*

$$\int_{\gamma} f(z) dz = 0$$

for every closed path γ in G . △

Proof. By Goursat's Lemma ([Theorem 3.1](#)), $\int_{\gamma} f(z) dz = 0$ for any triangle γ in G . By [Lemma 3.5](#), f has an antiderivative F in G . Now the Fundamental Theorem of Calculus ([Theorem 2.9](#)) gives the desired statement. □

Cauchy's Integral Theorem has different formulations and is valid under less restrictive conditions. We do not go into details here. However, it is important to note that it is only true if the region G does not have "holes". In this connection, recall that if γ is the unit circle and $f(z) = \frac{1}{z}$, then $\int_{\gamma} \frac{1}{z} = 2\pi i$ (see [Example 2.3](#)).

Chapter 4

Consequences of Cauchy's integral theorem

4.1 Cauchy's integral formula

We say that a simple closed path γ is *positively oriented*, if, as t increases, $\gamma(t)$ moves anticlockwise with respect to any point in the set enclosed by γ .

For $z_0 \in \mathbb{C}$ and $r > 0$, we denote by

$$\overline{D}(z_0; r) = \{z \in \mathbb{C} : |z - z_0| \leq r\}$$

the *closed disc* with center z_0 and radius r .

Theorem 4.1 (*Cauchy's Integral Formula*) Let G be a region, let $z_0 \in G$, and let $\overline{D}(z_0; r) \subseteq G$. Let γ be the circle that forms the boundary of $\overline{D}(z_0; r)$, with the positive orientation. Let f be a holomorphic function on G . Then

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw \quad \text{for any } z \in D(z_0; r).$$

△

For the proof of this theorem, we will need a version of Cauchy's Integral Theorem for functions that are holomorphic everywhere in the domain except for one point. We start with the corresponding version of Goursat's Lemma.

Lemma 4.2 Let $G \subseteq \mathbb{C}$ be an open set, Δ a closed triangle in G , and let γ denote the path along the edges on Δ . Suppose that f is continuous on G and holomorphic on $G \setminus \{z_0\}$, where $z_0 \in \Delta$. Then

$$\int_{\gamma} f(z) dz = 0.$$

△

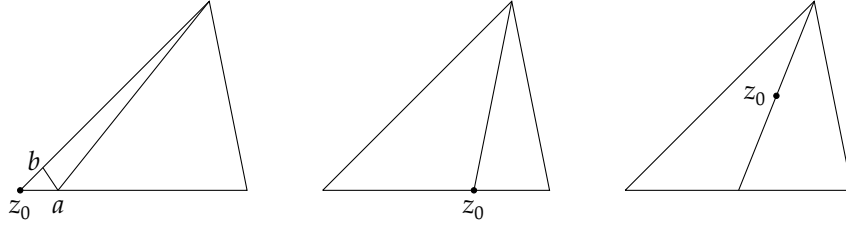


Figure 4.1: Subdividing triangles

Proof. We consider three cases.

A. First assume that z_0 is one of the vertices of Δ . By assumption, f is continuous on Δ , and Δ is a compact set. It follows that f is bounded on Δ . Let $M > 0$ be a constant such that $|f(z)| \leq M$ for all $z \in \Delta$. Let $\varepsilon > 0$ be an arbitrary positive number. We subdivide Δ into three triangles as shown in the left part of Figure 4.1. We denote the small triangle with the vertex z_0 by Δ_1 , and the other two triangles by Δ_2, Δ_3 ; γ_k with $k = 1, 2, 3$ denotes the path along the sides of Δ_k . We choose the points a and b such that $\ell(\gamma_1) < \varepsilon$. By Proposition 2.4,

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz + \int_{\gamma_3} f(z) dz.$$

For the triangles Δ_2 and Δ_3 , we can find open sets that contain Δ_2, Δ_3 , respectively, where f is holomorphic. By Goursat's Lemma (Theorem 3.1),

$$\int_{\gamma_2} f(z) dz = \int_{\gamma_3} f(z) dz = 0.$$

Using Corollary 2.7, we estimate

$$\left| \int_{\gamma_1} f(z) dz \right| \leq M \ell(\gamma_1) < M\varepsilon.$$

Altogether we obtain

$$\left| \int_{\gamma} f(z) dz \right| = \left| \int_{\gamma_1} f(z) dz \right| \leq M\varepsilon.$$

Since this is true for any $\varepsilon > 0$, we have $\int_{\gamma} f(z) dz = 0$.

b. Now assume that z_0 lies on an edge of Δ , but is not a vertex. We subdivide Δ into two triangles Δ_1, Δ_2 as shown in the middle part of **Figure 4.1**. Now part A is applicable to both Δ_1 and Δ_2 . Therefore,

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz = 0 + 0 = 0.$$

c. Finally, let z_0 is in the interior of Δ . We subdivide Δ into two triangles Δ_1, Δ_2 as shown in the right part of **Figure 4.1**. Now part B is applicable to both Δ_1 and Δ_2 . Therefore,

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz = 0 + 0 = 0.$$

□

Corollary 4.3 Let $G \subseteq \mathbb{C}$ be a star-shaped region. Let γ be a closed path in G . Suppose that f is continuous on G and holomorphic on $G \setminus \{z_0\}$, where $z_0 \in G$. Then

$$\int_{\gamma} f(z) dz = 0.$$

△

Proof. Apply **Lemma 4.2**, **Lemma 3.5** and **Theorem 2.9** exactly like in the proof of **Theorem 3.6**.

□

Proof of Theorem 4.1. Since $\overline{D}(z_0; r)$ is a compact set, we can find $\varepsilon > 0$ such that the open disc $U = D(z_0; r + \varepsilon)$ is contained in G . The open disc U is a convex region, and in particular is it star-shaped. The function f is holomorphic on U , and γ is a closed path in U .

For a fixed $z \in D(z_0; r)$, consider the function

$$g(w) = \begin{cases} \frac{f(w)-f(z)}{w-z}, & \text{if } w \neq z, \\ f'(z), & \text{if } w = z. \end{cases}$$

It is holomorphic on $U \setminus \{z\}$ and continuous on U . By **Corollary 4.3**,

$$0 = \int_{\gamma} g(w) dw = \int_{\gamma} \frac{f(w)}{w-z} dw - f(z) \int_{\gamma} \frac{1}{w-z} dw.$$

Hence,

$$f(z) = \left(\int_{\gamma} \frac{1}{w-z} dw \right)^{-1} \int_{\gamma} \frac{f(w)}{w-z} dw.$$

The statement of the theorem will follow if we show that

$$\int_{\gamma} \frac{1}{w-z} dw = 2\pi i.$$

To prove the latter formula, we differentiate the integral with respect to z . We claim that

$$\frac{\partial}{\partial z} \int_{\gamma} \frac{1}{w-z} dw = \int_{\gamma} \frac{1}{(w-z)^2} dw.$$

Indeed, put

$$\varphi(z) = \int_{\gamma} \frac{1}{w-z} dw.$$

Put $d = \min_{w \in \gamma} |w-z|$. Notice that $d > 0$. Take $h \in \mathbb{C}$ with $|h| < \frac{d}{2}$, then $z+h \in D(z_0; r)$, and

$$\frac{1}{h}(\varphi(z+h) - \varphi(z)) = \frac{1}{h} \int_{\gamma} \left(\frac{1}{w-z-h} - \frac{1}{w-z} \right) dw = \int_{\gamma} \frac{1}{(w-z-h)(w-z)} dw.$$

Since $|h| < \frac{1}{2}d$, we have that $|w-(z+h)| \geq |w-z| - |h| > \frac{1}{2}d$ for all $w \in \gamma$. We have

$$\begin{aligned} & \left| \frac{1}{h}(\varphi(z+h) - \varphi(z)) - \int_{\gamma} \frac{1}{(w-z)^2} dw \right| \\ &= \left| \int_{\gamma} \left(\frac{1}{(w-z-h)(w-z)} - \frac{1}{(w-z)^2} \right) dw \right| \\ &= \left| \int_{\gamma} \frac{h}{(w-z)^2(w-z-h)} dw \right| \\ &\leq \frac{2}{d^3} |h| \ell(\gamma) \rightarrow 0, \quad h \rightarrow 0, \end{aligned}$$

as desired. Notice that we used [Corollary 2.7](#) in the above estimate.

Now

$$\int_{\gamma} \frac{1}{(w-z)^2} dw = 0$$

by the Fundamental Theorem of Calculus (**Theorem 2.9**), because $\frac{1}{(w-z)^2}$ has an antiderivative $\frac{1}{w-z}$ on $U \setminus \{z\}$. We obtain that

$$\frac{\partial}{\partial z} \int_{\gamma} \frac{1}{w-z} dw = 0 \quad \text{for any } z \in D(z_0; r).$$

By **Theorem 1.14**, the function $z \mapsto \int_{\gamma} \frac{1}{w-z} dw$ is constant on $D(z_0; r)$. We already know from **Example 2.3** that

$$\int_{\gamma} \frac{1}{w-z_0} dw = 2\pi i.$$

It follows that

$$\int_{\gamma} \frac{1}{w-z} dw = 2\pi i \quad \text{for any } z \in D(z_0; r),$$

which finishes the proof of the theorem. □

4.2 Cauchy's integral formula for derivatives

Theorem 4.4 *Let G be a region, let $z_0 \in G$, and let $\overline{D}(z_0; r) \subseteq G$. Let γ be the circle that forms the boundary of $\overline{D}(z_0; r)$, with the positive orientation. Let f be a holomorphic function on G . Then $f^{(n)}(z)$ exists for any $z \in D(z_0; r)$, and*

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w-z)^{n+1}} dw, \quad n = 0, 1, 2, \dots$$

△

Proof. We prove the statement by induction. For $n = 0$ it is exactly the statement of the Cauchy Integral Formula (**Theorem 4.1**). For the induction step, assume that the formula holds for some $n = m$. Let $z \in D(z_0; r)$. The function f is continuous and thus bounded on $\overline{D}(z_0; r)$: there exists $M > 0$ such that $|f(z)| \leq M$ for all $z \in \overline{D}(z_0; r)$. With $d = \min_{w \in \gamma} |w - z| > 0$ and $|h| < \frac{1}{2}d$ we have

$$\begin{aligned} \frac{1}{h} (f^{(m)}(z+h) - f^{(m)}(z)) &= \frac{m!}{2\pi i h} \int_{\gamma} f(w) \left(\frac{1}{(w-z-h)^{m+1}} - \frac{1}{(w-z)^{m+1}} \right) dw \\ &= \frac{(m+1)!}{2\pi i h} \int_{\gamma} f(w) \left(\int_{[z, z+h]} \frac{1}{(w-\zeta)^{m+2}} d\zeta \right) dw. \end{aligned}$$

Hence,

$$\begin{aligned}
 & \frac{1}{h} (f^{(m)}(z+h) - f^{(m)}(z)) - \frac{(m+1)!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w-z)^{m+2}} dw \\
 &= \frac{(m+1)!}{2\pi i h} \int_{\gamma} f(w) \left(\int_{[z, z+h]} \left\{ \frac{1}{(w-\zeta)^{m+2}} - \frac{1}{(w-z)^{m+2}} \right\} d\zeta \right) dw \\
 &= \frac{(m+2)!}{2\pi i h} \int_{\gamma} f(w) \left(\int_{[z, z+h]} \left\{ \int_{[z, \zeta]} \frac{1}{(w-\tau)^{m+3}} d\tau \right\} d\zeta \right) dw.
 \end{aligned}$$

Notice that $|w - \zeta| > \frac{1}{2}d$ for all $w \in \gamma$, $\zeta \in [z, z+h]$, and $|w - \tau| > \frac{1}{2}d$ for all $w \in \gamma$, $\tau \in [z, \zeta]$. Thus,

$$\begin{aligned}
 & \left| \frac{1}{h} (f^{(m)}(z+h) - f^{(m)}(z)) - \frac{(m+1)!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w-z)^{m+2}} dw \right| \\
 & \leq \frac{(m+2)!}{2\pi|h|} \frac{M|h|^2 2^{m+3}}{d^{m+3}} \ell(\gamma) \rightarrow 0, \quad h \rightarrow 0,
 \end{aligned}$$

which proves the statement. $\square$

Corollary 4.5 *If f is holomorphic in a open set G , then f has derivatives of all orders in G , and all derivatives are holomorphic in G .* $\triangle$

Notice a huge difference between complex differentiability and real differentiability! A real function that is differentiable on a open interval may have a derivative that it not even continuous on that interval. A complex holomorphic function always has a derivative that is itself holomorphic.

Corollary 4.5 allows us to prove a converse of Goursat's Lemma (**Theorem 3.1**).

Theorem 4.6 (Morera's Theorem) *Suppose that f is continuous in an open set $G \subseteq \mathbb{C}$ and that $\int_{\gamma} f(z) dz = 0$ for all triangles γ in G . Then f is holomorphic in G .* $\triangle$

Proof. Let $z \in G$. Since G is open, we can find $r > 0$ such that $D(z; r) \subseteq G$. The disc $D(z; r)$ is a star-shaped region, therefore by **Lemma 3.5** f has an antiderivative F in $D(z; r)$. By **Corollary 4.5**, $f = F'$ is holomorphic in $D(z; r)$. Since z can be taken arbitrarily, f is holomorphic in G . $\square$

4.3 Power series representations of holomorphic functions

We have seen in [Theorem 1.31](#) that a convergent power series defines a holomorphic function. In this section we will prove the converse statement.

Theorem 4.7 (Taylor's Theorem) *Let f be a holomorphic function in $D(z_0; R)$, $R > 0$. Then there exist unique numbers $a_n \in \mathbb{C}$, $n = 0, 1, 2, \dots$, such that*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad z \in D(z_0; R).$$

The numbers a_n are given by

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw = \frac{f^{(n)}(z_0)}{n!},$$

where γ is a circle with center z_0 , radius r ($0 < r < R$) and the positive orientation. $\triangle$

Proof. Fix $z \in D(z_0; R)$ and take r such that $|z - z_0| < r < R$. Take γ be a circle with center z_0 and radius r . By Cauchy's integral formula ([Theorem 4.1](#)),

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw.$$

Notice that $|z - z_0| < r = |w - z_0|$ for any $w \in \gamma$, thus

$$\frac{1}{w - z} = \frac{1}{w - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{w - z_0}} = \frac{1}{w - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{w - z_0} \right)^n,$$

where the geometric series on the right-hand side converges. This gives us

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(w - z_0)^{n+1}} f(w) dw.$$

The function f is bounded on the compact set γ , i.e., there is $M > 0$ such that $|f(w)| \leq M$ for all $w \in \gamma$. We have

$$\left| \frac{(z - z_0)^n}{(w - z_0)^{n+1}} f(w) \right| \leq \frac{M}{r} \left(\frac{|z - z_0|}{r} \right)^n = M_n.$$

The series $\sum_{n=0}^{\infty} M_n$ converges. This justifies the interchange of integration and summation.¹ Thus,

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n$$

as desired. **Theorem 4.4** gives the formula $a_n = \frac{f^n(z_0)}{n!}$. By **Corollary 1.32**, the coefficients a_n are uniquely defined. □

Notice that **Theorem 1.31** and **Theorem 4.7** together establish the result that a complex function is holomorphic in an open set G if and only if it is analytic.

Corollary 4.8 (Cauchy's Inequalities) Assume that $|f(z)| \leq M$ for all z on the circle with center z_0 and radius r . Then the coefficients of Taylor's series $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ satisfy *Cauchy's inequalities*

$$|a_n| \leq \frac{M}{r^n}, \quad n = 0, 1, 2, \dots$$

△

Proof. By **Corollary 2.7** we have

$$|a_n| = \left| \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw \right| \leq \frac{1}{2\pi} \frac{M}{r^{n+1}} 2\pi r = \frac{M}{r^n}.$$

□

Corollary 4.8 says that the coefficients of Taylor's series of f are dominated by the geometric progression $\left(\frac{M}{r^n}\right)_{n=0}^{\infty}$.

4.4 Liouville's theorem and the fundamental theorem of algebra

Definition 4.9 A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is called an *entire function* if it is holomorphic on $\mathbb{C}$. △

¹This is a known statement and will be proven in the tutorial: Let $G \subseteq \mathbb{C}$ be a region, let γ be a path in G , and let $(f_n)_{n=1}^{\infty}$ be a sequence of continuous functions $f_n : G \rightarrow \mathbb{C}$. Suppose that for each $n \in \mathbb{N}$ there is $M_n \geq 0$ such that $|f_n(z)| \leq M_n$ for all $z \in \gamma$, and that the series $\sum_{n=1}^{\infty} M_n$ converges. Then $\sum_{n=1}^{\infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} \sum_{n=1}^{\infty} f_n(z) dz$.

Polynomials, the exponential function, the sine and the cosine functions are examples of entire functions. On the other hand, the functions $\frac{1}{z}$, $\tan z$ are not entire.

Notice that an entire function can be represented by a power series $\sum_{n=0}^{\infty} a_n z^n$ that converges on the entire complex plane. In some sense, entire functions can be considered as a generalization of polynomials.

Lemma 4.10 *Let f be an entire function. Assume that there are constants $n \in \mathbb{N} \cup \{0\}$, $M, R > 0$ such that*

$$|f(z)| \leq M|z|^n \quad \text{for all } |z| \geq R.$$

Then f is a polynomial of degree at most n .

△

Proof. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$. Consider the circle $\gamma(t) = re^{it}$, $t \in [0, 2\pi]$, where $r > R$. Then $|f(z)| \leq Mr^n$ on this circle. By **Corollary 4.8**,

$$|a_k| \leq \frac{Mr^n}{r^k} = Mr^{n-k}, \quad k \in \mathbb{N} \cup \{0\}.$$

If $k > n$, then $r^{n-k} \rightarrow 0$ as $r \rightarrow \infty$. Thus, $a_k = 0$ for $k > n$.

□

The next result is known as Liouville's Theorem. Recall that a function $f : S \rightarrow \mathbb{C}$ is called **bounded** on S , if there is a constant $M > 0$ such that $|f(z)| \leq M$ for all $z \in S$.

Theorem 4.11 (Liouville's Theorem) *An entire function that is bounded on $\mathbb{C}$ is constant.*

△

Proof. Apply **Lemma 4.10** with $n = 0$.

□

Notice that Liouville's Theorem gives another proof of the fact that the functions $\sin z$ and $\cos z$ are not bounded on $\mathbb{C}$: they are entire, but not constant.

Liouville's Theorem provides a simple proof of the Fundamental Theorem of Algebra.

Theorem 4.12 (Fundamental Theorem of Algebra) *Let p be a non-constant polynomial with complex coefficients. Then p has a zero in $\mathbb{C}$.*

△

Proof. Suppose that p is a non-constant polynomial with complex coefficients and no zeros. Then the function $\frac{1}{p}$ is defined and holomorphic on the entire complex plane $\mathbb{C}$. We have $|p(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$. Therefore, there exists $R > 0$ such that $|p(z)| \geq 1$ for all $|z| \geq R$. Hence, $\left| \frac{1}{p(z)} \right| \leq 1$ for all $|z| \geq R$. On the other hand, the function $\frac{1}{p}$ is continuous and therefore bounded on the closed disc $\{z \in \mathbb{C} : |z| \leq R\}$. We obtain that $\frac{1}{p}$ is bounded on $\mathbb{C}$. By Liouville's Theorem ([Theorem 4.11](#)), $\frac{1}{p}$ is constant. This implies that p is constant, which is a contradiction. $\square$

4.5 Uniform convergence of holomorphic functions

We know that a uniform limit of a sequence of continuous functions is continuous. We will now show that a uniform limit of a sequence of holomorphic functions is holomorphic.

Theorem 4.13 *Let $(f_n)_{n=1}^\infty$ be a sequence of functions holomorphic in a region $G \subseteq \mathbb{C}$ that converges to a function $f : G \rightarrow \mathbb{C}$, and the convergence is uniform on any compact subset of G . Then f is holomorphic in G , the sequence of derivatives $(f_n^{(k)})_{n=1}^\infty$ converges to $f^{(k)}$ for any $k \in \mathbb{N}$, and for every $z \in G$ there is a closed disc $\overline{D}(z; r) \subseteq G$ with some $r > 0$ such that the convergence is uniform in $\overline{D}(z; r) \subseteq G$.* $\triangle$

Remark This type of convergence is called *locally uniform convergence*. $\triangle$

Proof. The function f is continuous as the limit of a uniformly convergent sequence. If Δ is a closed triangle in G and η is the path along its edges, then

$$\int_{\eta} f(z) dz = \int_{\eta} \lim_{n \rightarrow \infty} f_n(z) dz = \lim_{n \rightarrow \infty} \int_{\eta} f_n(z) dz = 0;$$

here we used the statement from the tutorial about uniform convergence to interchange the limit and the integration,² and Goursat's Lemma ([Theorem 3.1](#))

²Let $G \subseteq \mathbb{C}$ be a region, let γ be a path in G , and let $(f_n)_{n=1}^\infty$ be a sequence of continuous functions $f_n : G \rightarrow \mathbb{C}$. Suppose that f_n converges to $f : \gamma \rightarrow \mathbb{C}$ uniformly on γ . Then f is continuous on γ and $\lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} f(z) dz$

for the holomorphic function f_n in the last step. But then by Morera's Theorem ([Theorem 4.6](#)) also f is holomorphic in G .

Let $z \in G$. Since G is open, we can find a closed disc $\overline{D}(z; R) \subseteq G$, $R > 0$. Let γ denote the circle with center z and radius R . Take any $0 < r < R$. For any $\zeta \in \overline{D}(z; r)$ and $w \in \gamma$ we have $|\zeta - w| \geq R - r > 0$. By Cauchy's integral formula for derivatives ([Theorem 4.4](#)),

$$\begin{aligned} |f_n^{(k)}(\zeta) - f^{(k)}(\zeta)| &= \left| \frac{k!}{2\pi i} \int_{\gamma} \frac{f_n(w) - f(w)}{(w - \zeta)^{k+1}} dw \right| \\ &\leq \frac{k!}{2\pi(R-r)^{k+1}} 2\pi R \sup_{w \in \gamma} |f_n(w) - f(w)|. \end{aligned}$$

Because of the uniform convergence of f_n to f on the compact set γ ,

$$\sup_{w \in \gamma} |f_n(w) - f(w)| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

It follows that $f_n^{(k)}$ converges to $f^{(k)}$ uniformly in $\overline{D}(z; r)$ for any $k \in \mathbb{N}$. □

[Theorem 4.13](#) demonstrates once again a difference between the complex and the real case: in the real case, a uniform limit of a sequence of differentiable functions need not be differentiable, and even if it is differentiable, one can interchange the limit and the differentiation only under some additional conditions.

Notice that [Theorem 4.13](#) gives another proof of the fact that the sum of a power series is holomorphic and that the power series can be differentiated term by term.

4.6 The coincidence principle

Let $S \subseteq \mathbb{C}$. Recall that $a \in \mathbb{C}$ is a *limit point* of S if its every neighborhood $D(a; \varepsilon)$, $\varepsilon > 0$, contains a point of $S \setminus \{a\}$. Limit points are also called *cluster points* or *accumulation points* of S . The set of all limit points of S is called the *derived set* of S and denoted as S' . A point $z \in S$ which is not a limit point of S is called an *isolated point* of S .

Lemma 4.14 Assume that f is holomorphic in an open disc $D(z_0; r)$ with some $z_0 \in \mathbb{C}$, $r > 0$. If z_0 is a limit point of the set $\{\zeta \in D(z_0; r) : f(\zeta) = 0\}$, then $f(z) = 0$ for all $z \in D(z_0; r)$. △

Proof. By continuity of f we have $f(z_0) = 0$. Let

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

be Taylor's series of f in $D(z_0; r)$. We have $a_0 = f(z_0) = 0$. If all coefficients $a_n = 0$, $n \in \mathbb{N}$, then $f \equiv 0$ in $D(z_0; r)$ and we are done.

Now assume that not all of the coefficients a_n are zero. Let $m \in \mathbb{N}$ be the smallest index with $a_m \neq 0$. Then

$$f(z) = (z - z_0)^m \sum_{k=0}^{\infty} a_{m+k}(z - z_0)^k.$$

The radius of convergence of the power series $\sum_{k=0}^{\infty} a_{m+k}(z - z_0)^k$ is the same as the radius of convergence of $\sum_{n=0}^{\infty} a_n(z - z_0)^n$; it is at least r . Consequently, $\sum_{k=0}^{\infty} a_{m+k}(z - z_0)^k$ converges to a holomorphic function in $D(z_0; r)$. Put

$$g(z) = \sum_{k=0}^{\infty} a_{m+k}(z - z_0)^k, \quad z \in D(z_0; r).$$

Since $g(z_0) = a_m \neq 0$ and g is continuous, there is $\varepsilon > 0$ such that $g(z) \neq 0$ for all $z \in D(z_0; \varepsilon)$. But this means that z_0 is an isolated point of $\{\zeta \in D(z_0; r) : f(\zeta) = 0\}$, which contradicts the assumption. It follows that $a_n = 0$, $n \in \mathbb{N}$, and therefore $f \equiv 0$ in $D(z_0; r)$ as desired. □

Theorem 4.15 (*The Identity Theorem*) Suppose that f is a holomorphic function in a region $G \subseteq \mathbb{C}$. If the set $\{\zeta \in G : f(\zeta) = 0\}$ has a limit point in G , then $f \equiv 0$ in G . △

Proof. Let $Z = \{\zeta \in G : f(\zeta) = 0\}$ and $L = Z' \cap G$ be the set of all limit points of Z in G . By the assumption, $L \neq \emptyset$. Since f is continuous, $L \subseteq Z$.

Take an arbitrary $z_0 \in L$. Since G is open, there is an open disc $D(z_0; r)$ with some $r > 0$ that it contained in G . By **Lemma 4.14**, $f(z) = 0$ for all $z \in D(z_0; r)$. This means that L is open.

On the other hand, if $z_0 \in G \setminus L$, then z_0 is not a limit point of Z , and thus there exists an open disc $D(z_0; r)$, $r > 0$, that does not contain points from Z except, perhaps, for z_0 . This means that also $G \setminus L$ is open.

Thus, we have a decomposition of the region G in the union $G = L \cup (G \setminus L)$ of two open disjoint sets. Since G is connected, this is only possible if $G \setminus L = \emptyset$. It follows that $G = L$, and in particular $f \equiv 0$ in G . □

Corollary 4.16 If f is holomorphic in a region G and not constant, then the set $\{\zeta \in G : f(\zeta) = 0\}$ has no limit points in G . △

Corollary 4.17 (The Coincidence Principle) Let f and g be holomorphic in a region G . If the set $\{\zeta \in G : f(\zeta) = g(\zeta)\}$ has a limit point in G , then $f \equiv g$ in G . △

Example 4.18 A. Let f be holomorphic in $\mathbb{C}$ and $f\left(\frac{1}{n}\right) = \sin\left(\frac{1}{n}\right)$, $n \in \mathbb{N}$. The set $\left\{\frac{1}{n} : n \in \mathbb{N}\right\}$ has a limit point 0 in $\mathbb{C}$. Thus, $f(z) = \sin z$ for all $z \in \mathbb{C}$.

B. Now assume that f is holomorphic in $\mathbb{C} \setminus \{0\}$ and that $f(z) = \sin\left(\frac{1}{z}\right)$ whenever $z = \frac{1}{\pi n}$, $n \in \mathbb{N}$. Then it does not follow that necessarily $f(z) = \sin\left(\frac{1}{z}\right)$. Indeed, also the function $f \equiv 0$ fits the conditions. The Coincidence Principle cannot be applied because the only limit point of the set $\left\{\frac{1}{\pi n} : n \in \mathbb{N}\right\}$ is zero, and it does not lie in the region $\mathbb{C} \setminus \{0\}$. △

The Coincidence Principle can be used to extend certain functional identities known for reals to the complex case. For example, we know that $\sin^2 z + \cos^2 z = 1$ if $z \in \mathbb{R}$. The function $z \mapsto \sin^2 z + \cos^2 z$ is an entire function, and the set $\mathbb{R}$ has a limit point in $\mathbb{C}$, and therefore the identity is valid for all $z \in \mathbb{C}$.

4.7 The maximum modulus principle

We finish this chapter with a short section where we present one further very important consequence of Cauchy's Integral formula — the Maximum Modulus Principle. The proofs will be given in the tutorial.

Theorem 4.19 (Gauss' Mean Value Theorem) Let f be a holomorphic function in the open disc $D(z; R)$ with some $z \in \mathbb{C}$, $R > 0$. Let $0 < r < R$. Then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(z + re^{i\theta}) d\theta.$$

△

Gauss' Mean Value Theorem says that the average value of the function f on the circle with center z is equal to the value of f at z .

For a set G , we denote by ∂G its boundary.

Theorem 4.20 (*The Maximum Modulus Principle*) Let G be a bounded region. Let f be holomorphic in G and continuous in $\overline{G}$. Then $|f|$ attains its maximum on the boundary of G . If f is non-constant, then

$$|f(z)| < \max_{w \in \partial G} |f(w)| \quad \text{for any } z \in G.$$

△

Chapter 5

Singularities and residues

5.1 Classification of singularities

Rational functions of functions like $\frac{\sin z}{z}$, $\tan z = \frac{\sin z}{\cos z}$, $\sin\left(\frac{1}{z}\right)$ etc. are holomorphic except for a finite or countable set, namely, the set of zeros of their denominators. We are now going to investigate such situations.

For $z_0 \in \mathbb{C}$ and $r > 0$ we define a *punctured disc*

$$D'(z_0; r) = \{z \in \mathbb{C} : 0 < |z - z_0| < r\}.$$

Definition 5.1 A point $z_0 \in \mathbb{C}$ is called an *isolated singularity* of f , if f is holomorphic in $D'(z_0; r)$ with some $r > 0$, but not at z_0 .

△

For example, the functions $\frac{1}{z}$, $\sin\left(\frac{1}{z}\right)$, $\frac{\sin z}{z}$ have an isolated singularity at $z_0 = 0$. However, there are differences in the behavior of the function f at z_0 : for instance, in our examples the limit $\lim_{z \rightarrow 0} f(z)$ does not exist for the functions $\frac{1}{z}$ and $\sin\left(\frac{1}{z}\right)$, but $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$. While $\lim_{z \rightarrow 0} \left|\frac{1}{z}\right| = \infty$, as similar statement does not hold for the function $\sin\left(\frac{1}{z}\right)$.

The above examples motivate us to introduce the following classification of isolated singularities.

Definition 5.2 Let z_0 be an isolated singularity of a function f .

- (1) z_0 is called a *removable singularity*, if $f(z)$ is bounded on $D'(z_0; r)$ with some $r > 0$.
- (2) z_0 is called a *pole* of f , if $\lim_{z \rightarrow z_0} |f(z)| = \infty$.
- (3) z_0 is called an *essential singularity*, if z_0 is an isolated singularity but neither a removable singularity nor a pole.

△

Here the limit $\lim_{z \rightarrow z_0} |f(z)| = \infty$ means that for any $M > 0$ there is $\delta > 0$ such that

$$|f(z)| > M \quad \text{for all } z \in D'(z_0; \delta).$$

Notice that z_0 is an essential singularity of f if and only if there exist two sequences $(z_n)_{n=1}^\infty, (w_n)_{n=1}^\infty$ in $D'(z_0; r)$ with $\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} w_n = z_0$ such that $(f(z_n))_{n=1}^\infty$ is bounded but $(f(w_n))_{n=1}^\infty$ is unbounded.

For example, the function $\frac{\sin z}{z}$ has a removable singularity at $z_0 = 0$. The function $\frac{1}{z}$ has a pole at $z_0 = 0$. The function $\sin\left(\frac{1}{z}\right)$ has an essential singularity at $z_0 = 0$. Indeed, if $z_n = \frac{1}{n}$ with $n \in \mathbb{N}$, then $f(z_n) = \sin n$, and the sequence $(f(z_n))_{n=1}^\infty$ is bounded. If $w_n = \frac{i}{n}$ with $n \in \mathbb{N}$, then $f(w_n) = \sin(-in) = \frac{1}{2i}(e^n - e^{-n})$, and the sequence $(f(w_n))_{n=1}^\infty$ is unbounded.

Let us consider one further example: $f(z) = \tan\left(\frac{1}{z}\right)$. The point $z_0 = 0$ is not an isolated singularity of $f(z)$. Indeed, every punctured disc $D'(0; r)$, $r > 0$, contains infinitely many zeros of the denominator $\cos\left(\frac{1}{z}\right)$. This is a so-called *non-isolated essential singularity*. We will not study them in this course.

5.2 Removable singularities, essential singularities, and poles

The next theorem will explain why singularities in **Definition 5.2** (1) are called removable.

Theorem 5.3 (*Riemann's Extension Theorem*) *If a function f is holomorphic and bounded in a punctured disc $D'(z_0; r)$ with some $z_0 \in \mathbb{C}$, $r > 0$, then the limit $L = \lim_{z \rightarrow z_0} f(z)$ exists. The new function*

$$\hat{f}(z) = \begin{cases} f(z), & z \in D'(z_0; r), \\ L, & z = z_0, \end{cases}$$

is holomorphic in $D(z_0; r)$.

△

Proof. Consider the function

$$g(z) = \begin{cases} (z - z_0)^2 f(z), & z \in D'(z_0; r), \\ 0, & z = z_0. \end{cases}$$

The function g is holomorphic in $D'(z_0; r)$ as a product of two holomorphic functions. At the point z_0 we have

$$g'(z_0) = \lim_{z \rightarrow z_0} \frac{g(z) - g(z_0)}{z - z_0} = \lim_{z \rightarrow z_0} (z - z_0)f(z) = 0.$$

Thus, g is holomorphic in the disc $D(z_0; r)$. Therefore, we can expand g in Taylor's series around z_0 . Since $g(z_0) = g'(z_0) = 0$, we have

$$g(z) = \sum_{n=2}^{\infty} \frac{g^{(n)}(z_0)}{n!} (z - z_0)^n = (z - z_0)^2 h(z),$$

where $h(z) = \sum_{n=0}^{\infty} \frac{g^{(n+2)}(z_0)}{(n+2)!} (z - z_0)^n$ is a function that is holomorphic in $D(z_0; r)$. Moreover, $f(z) = h(z)$ for all $z \in D'(z_0; r)$. It follows that

$$\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow z_0} h(z) = h(z_0).$$

If we put $L = h(z_0)$, then the function $\hat{f}$ defined above coincides with h and thus it is holomorphic in $D(z_0; r)$. □

Remark Clearly, if the limit $\lim_{z \rightarrow z_0} f(z)$ exists, then f is bounded in some neighborhood $D'(z_0; r)$, $r > 0$. Thus, z_0 is a removable singularity of f if and only if the limit $\lim_{z \rightarrow z_0} f(z)$ exists. △

With the help of Riemann's Extension Theorem (**Theorem 5.3**), we can prove an interesting property of essential singularities.

Theorem 5.4 (The Casorati-Weierstrass Theorem) Suppose that f is holomorphic in $D'(z_0; R)$ with some $R > 0$ and has an essential singularity at z_0 . Then for any number $w_0 \in \mathbb{C}$ there exists a sequence $(z_n)_{n=1}^{\infty}$ such that $\lim_{n \rightarrow \infty} z_n = z_0$ and $\lim_{n \rightarrow \infty} f(z_n) = w_0$. △

Proof. We will prove the theorem by contradiction. Suppose that there exist $w_0 \in \mathbb{C}$, $0 < r < R$ and $\varepsilon > 0$ such that

$$0 < |z - z_0| < r \implies |f(z) - w_0| \geq \varepsilon.$$

(Make sure that you understand where this statement comes from.)

Consider the function

$$g(z) = \frac{1}{f(z) - w_0}.$$

By our assumption, g is holomorphic and bounded in $D'(z_0; r)$. By Riemann's Extension Theorem (**Theorem 5.3**), the limit $\lim_{z \rightarrow z_0} g(z) = L$ exists.

Now if $L = 0$, then $\lim_{z \rightarrow z_0} |f(z)| = \infty$, so that z_0 is a pole. If $L \neq 0$, then $\lim_{z \rightarrow z_0} f(z) = w_0 + \frac{1}{L}$, and z_0 is a removable singularity. In both cases z_0 is not an essential singularity, which is a contradiction. $\square$

Remark In fact, much more is true in a neighborhood of an essential singularity: Great Picard's Theorem says that if f is holomorphic in $D'(z_0; R)$ and has an essential singularity at z_0 , then any value $w_0 \in \mathbb{C}$, with possibly one exception, is attained infinitely often in $D'(z_0; R)$. We will not be able to prove this theorem in our course.

As an example, consider the function $\exp\left(\frac{1}{z}\right)$. It has an essential singularity at $z_0 = 0$. The exceptional value that this function does not attain is $w_0 = 0$. $\triangle$

Now let us turn our attention to the poles. Suppose that f has a pole at z_0 , i.e., $\lim_{z \rightarrow z_0} |f(z)| = \infty$. The following is true.

Proposition 5.5 *Suppose that f is holomorphic in a punctured disc $D'(z_0; R)$ with some $R > 0$, and z_0 is a pole of f . Then there exists a natural number $n \in \mathbb{N}$ and a function φ that is holomorphic in $D(z_0; r)$ with some $0 < r \leq R$ such that $\varphi(z_0) \neq 0$ and*

$$f(z) = \frac{\varphi(z)}{(z - z_0)^n}, \quad z \in D'(z_0; r).$$

The number $n \in \mathbb{N}$ and the function φ as above are uniquely determined. $\triangle$

Proof. Suppose that f is holomorphic in some punctured disc $D'(z_0; R)$ and z_0 is a pole of f . Since $\lim_{z \rightarrow z_0} |f(z)| = \infty$, there exists $0 < r \leq R$ such that $|f(z)| \geq 1$

for all $z \in D'(z_0; r)$. Then $\frac{1}{|f(z)|} \leq 1$ in $D'(z_0; r)$, i.e. the function $\frac{1}{f}$ is bounded in $D'(z_0; r)$. It follows that $\frac{1}{f}$ has a removable singularity at z_0 with $\lim_{z \rightarrow z_0} \frac{1}{f(z)} = 0$.

Consider the function

$$g(z) = \begin{cases} \frac{1}{f(z)}, & z \in D'(z_0; r), \\ 0, & z = z_0. \end{cases}$$

By above, g is holomorphic in $D(z_0; r)$. We consider its Taylor expansion

$$g(z) = \sum_{k=1}^{\infty} c_k (z - z_0)^k.$$

Let n be the smallest natural number with $c_n \neq 0$. Then $g(z) = (z - z_0)^n h(z)$, where $h(z) = \sum_{k=0}^{\infty} c_{k+n} (z - z_0)^k$ is a holomorphic function in $D(z_0; r)$ and $h(z) \neq 0$, $z \in D(z_0; r)$. Therefore, the function $\varphi(z) = \frac{1}{h(z)}$ is also holomorphic in $D(z_0; r)$. For the function f we have that

$$f(z) = \frac{\varphi(z)}{(z - z_0)^n}, \quad \text{where } \varphi(z_0) \neq 0.$$

To show the uniqueness, suppose that

$$f(z) = \frac{\varphi_1(z)}{(z - z_0)^{n_1}} = \frac{\varphi_2(z)}{(z - z_0)^{n_2}}$$

are two such representations. Assume that $n_1 \neq n_2$. We may assume without loss of generality that $n_2 > n_1$. Then

$$(z - z_0)^{n_2 - n_1} \varphi_1(z) = \varphi_2(z).$$

But the left-hand side has value 0 for $z = z_0$, while the right-hand side is $\varphi_2(z_0) \neq 0$. This is impossible, and we conclude that $n_1 = n_2$. But we also necessarily have $\varphi_1 \equiv \varphi_2$.

□

Definition 5.6 The number $n \in \mathbb{N}$ from **Proposition 5.5** is called an *order of the pole* z_0 . Equivalently, n is the smallest natural number such that $(z - z_0)^n f(z)$ is bounded in $D'(z_0; R)$.

If $n = 1$, then z_0 is called a *simple pole*. If $n = 2$, then z_0 is called a *double pole*, and so on.

△

Example 5.7 The function $f(z) = \frac{1}{z^2(z^2+1)}$ has a double pole at $z = 0$ and simple poles at $z = \pm i$. △

Definition 5.8 A function f is called *meromorphic* in a region $G \subseteq \mathbb{C}$ if f is holomorphic in G except possibly for poles. △

Since poles are isolated singularities, there are at most countably many of them. Examples of meromorphic functions are rational functions or the function $\tan z$. More generally, the quotients $\frac{f}{g}$ of two holomorphic functions f, g with $g \not\equiv 0$ are meromorphic. Indeed, we know from the Identity Theorem ([Theorem 4.15](#)) that the set of zeros of g is at most countable and has no limit points. This set is the set of poles of $\frac{f}{g}$. Also the converse statement is true: any meromorphic functions can be represented locally (i.e., in a neighborhood of a given point) as a quotient of two holomorphic functions. We will not discuss this result in our course.

Meromorphic functions can be regarded as a generalization of rational functions in the same way like entire functions can be regarded as a generalization of polynomials.

5.3 Residues

Definition 5.9 Let f be holomorphic in a region G except possibly for isolated singularities. The *residue* of f at a point $z_0 \in G$ is the number

$$\text{Res}(f, z_0) = \frac{1}{2\pi i} \int_{\gamma} f(z) dz,$$

where $\gamma(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, and $r > 0$ is chosen so that the closed disc $\overline{D}(z_0; r)$ lies in G and contains no singularities of f except possibly for z_0 . △

Clearly, if f is holomorphic at z_0 , or if z_0 is a removable singularity of f , then $\text{Res}(f, z_0) = 0$.

Example 5.10 Consider the function $f(z) = \frac{1}{z-a}$, $a \in \mathbb{C}$. By [Example 2.3](#) we have $\text{Res}(f, a) = 1$. △

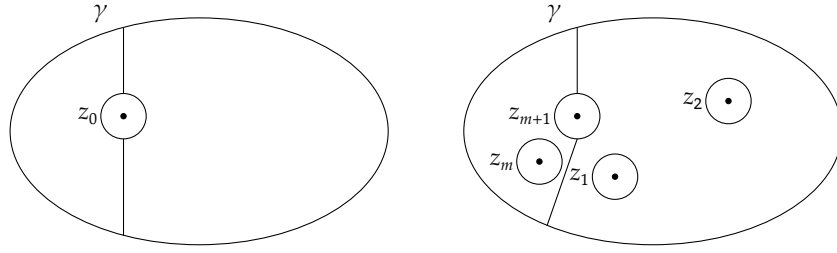


Figure 5.1: Proof of the Residue Theorem

Theorem 5.11 (The Residue Theorem) Let f be holomorphic in $\mathbb{C}$ except possibly for isolated singularities. Let γ be a positively oriented simple closed path in $\mathbb{C}$, and no singularities lie on γ . Let $z_1, z_2, \dots, z_n$ be singularities of f that lie inside γ . Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k).$$

△

Idea of the proof. Notice that any disc of finite radius contains finitely many isolated points of f .

We use induction in n .

First assume that f has only one singularity z_1 inside γ . We put a small circle γ_1 around z_1 and add two line segments connecting γ_1 to γ , see the left part of Figure 5.1. Now denote by η_1 the positively oriented path that includes the right part of γ , the line segments and the right half of the circle γ_1 , and by η_2 the positively oriented path that includes the left part of γ , the line segments and the left half of the circle γ_1 . By Proposition 2.4,

$$\int_{\gamma} f(z) dz = \int_{\eta_1} f(z) dz + \int_{\gamma_1} f(z) dz + \int_{\eta_2} f(z) dz.$$

On the other hand, both paths η_1 and η_2 can be embedded into star-shaped domains where f is holomorphic. Thus, by Cauchy's Integral Theorem (Theorem 3.6), $\int_{\eta_1} f(z) dz = \int_{\eta_2} f(z) dz = 0$. We obtain that

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz = 2\pi i \text{Res}(f, z_1).$$

For the induction step, assume that we have proven the formula for $n = 1, \dots, m$ with some $m \in \mathbb{N}$. Assume that f has $m + 1$ singularities $z_1, \dots, z_{m+1}$

inside γ . Put small circles around the singularities. Take one of them (say, z_{m+1}) and add line segments connecting the corresponding small circle to γ (see the right part of **Figure 5.1**). Again,

$$\int_{\gamma} f(z) dz = \int_{\eta_1} f(z) dz + \int_{\gamma_1} f(z) dz + \int_{\eta_2} f(z) dz,$$

where by the induction hypothesis $\int_{\eta_1} f(z) dz$ is equal to $2\pi i$ times the sum of residues at singularities that lie inside η_1 , $\int_{\eta_2} f(z) dz$ is equal to $2\pi i$ times the sum of residues at singularities that lie inside η_2 , and $\int_{\gamma_1} f(z) dz$ gives again $2\pi i \operatorname{Res}(f, z_{m+1})$. The desired formula follows. $\square$

Theorem 5.11 is a powerful method for calculation of integrals of meromorphic functions. In the following statement we present rules for computing residues at poles. Finding residues at essential singularities is more difficult.

Theorem 5.12 *Let f be holomorphic in $D'(z_0; r)$ with some $r > 0$, and let z_0 be a pole of f .*

(1) *If z_0 is a simple pole of f , then*

$$\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z).$$

(2) *If $f(z) = \frac{g(z)}{h(z)}$, where the functions g and h are holomorphic in $D(z_0; r)$, $g(z_0) \neq 0$, $h(z_0) = 0$ and $h'(z_0) \neq 0$ (i.e., h has a simple zero at z_0), then*

$$\operatorname{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}.$$

(3) *If z_0 is a pole of order $n > 1$ of f , then*

$$\operatorname{Res}(f, z_0) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} \left((z - z_0)^n f(z) \right) \Big|_{z=z_0}.$$

$\triangle$

Proof. (1) Suppose that f has a simple pole at z_0 . By **Proposition 5.5**,

$$f(z) = \frac{\varphi(z)}{z - z_0},$$

where the function φ is holomorphic in $D(z_0; r')$ with some $0 < r' \leq r$ and $\varphi(z_0) \neq 0$. Let γ be a small circle around z_0 . By Cauchy's Integral Formula (**Theorem 4.1**),

$$\text{Res}(f, z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{\varphi(z)}{z - z_0} dz = \varphi(z_0) = \lim_{z \rightarrow z_0} \varphi(z) = \lim_{z \rightarrow z_0} (z - z_0)f(z).$$

(2) By (1) and taking into account that $h(z_0) = 0$ we have

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) \frac{g(z)}{h(z)} = \lim_{z \rightarrow z_0} g(z) \cdot \lim_{z \rightarrow z_0} \frac{z - z_0}{h(z) - h(z_0)} = \frac{g(z_0)}{h'(z_0)}.$$

(3) In this case

$$f(z) = \frac{\varphi(z)}{(z - z_0)^n},$$

where φ is holomorphic in $D(z_0; r')$ with some $0 < r' \leq r$ and $\varphi(z_0) \neq 0$. Let γ be a small circle around z_0 . By Cauchy's Integral Formula for derivatives (**Theorem 4.4**),

$$\begin{aligned} \text{Res}(f, z_0) &= \frac{1}{2\pi i} \int_{\gamma} \frac{\varphi(z)}{(z - z_0)^n} dz = \frac{\varphi^{(n-1)}(z_0)}{(n-1)!} \\ &= \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} \left((z - z_0)^n f(z) \right) \Big|_{z=z_0}. \end{aligned}$$

□

Remark Notice that $\text{Res}(f, z_0)$ is the coefficient of $(z - z_0)^{n-1}$ in the Taylor expansion of φ about z_0 .

△

Example 5.13 The function $f(z) = \frac{e^z}{(z-2)^3}$ has a triple point at $z_0 = 2$. Here $n = 3$ and $\varphi(z) = e^z$. We have

$$\text{Res}(f, 2) = \frac{1}{2!} (e^z)'' \Big|_{z=2} = \frac{1}{2} e^2.$$

Alternatively, we use the Taylor expansion of $\varphi(z) = e^z$ about $z_0 = 2$:

$$e^z = e^2 e^{z-2} = e^2 \sum_{k=0}^{\infty} \frac{(z-2)^k}{k!}.$$

The residue $\text{Res}(f, 2)$ is the coefficient with $k = 2$, i.e.

$$\text{Res}(f, 2) = e^2 \frac{1}{2!} = \frac{1}{2} e^2.$$

△

5.4 Residues at infinity

For the analysis of the behavior of holomorphic functions as $|z| \rightarrow \infty$ we introduce the *extended complex plane* $\overline{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ with the following topology. We define an open disc with radius $r > 0$ at infinity as

$$D(\infty; r) = \{z \in \mathbb{C} : |z| > r\} \cup \{\infty\},$$

and the punctured open disc as

$$D'(\infty; r) = \{z \in \mathbb{C} : |z| > r\}.$$

A set $G \subseteq \overline{\mathbb{C}}$ is called *open* if for any $z \in G$ there is $r > 0$ such that $D(z; r) \subseteq G$. The set $\overline{\mathbb{C}}$ has a geometric interpretation. Put a sphere that touches the plane at the point $(0, 0)$. Then the north pole of the sphere corresponds to ∞ . For any point z on the plane, we associate to it the point on the sphere where the line segment connecting z with the north pole intersects the sphere. This gives a one-to-one correspondence between points on the plane and points on the sphere excluding the north pole. We do not go into details.

To a function f defined on $D(\infty; R)$ we associate the function $\tilde{f}$ on $D\left(0; \frac{1}{R}\right)$ defined by

$$\tilde{f}(z) = \begin{cases} f\left(\frac{1}{z}\right), & 0 < |z| < \frac{1}{R}, \\ f(\infty), & z = 0. \end{cases}$$

We say that f is continuous at ∞ if $\tilde{f}$ is continuous at 0. Similarly, we say that f is differentiable at ∞ if $\tilde{f}$ is differentiable at 0. We use this idea also to define singularities of f at ∞ .

Definition 5.14 Let f be a function holomorphic in $D'(\infty; R)$ with some $R > 0$. We say that f has a *removable singularity* at ∞ if $\tilde{f}(z) = f\left(\frac{1}{z}\right)$ has a removable singularity at 0. We say that f has a *pole of order n* at ∞ if $\tilde{f}$ has a pole of order n at 0. We say that f has an *essential singularity* at ∞ if $\tilde{f}$ has an essential singularity at 0.

△

For example, the function z^3 has a triple pole at ∞ , and the function $\frac{z+1}{z}$ has a removable singularity at ∞ .

Definition 5.15 Let f be a function holomorphic in $D'(\infty; R)$ with some $R > 0$. The *residue* at ∞ of the function f is defined as

$$\text{Res}(f, \infty) = \frac{1}{2\pi i} \int_{\gamma} f(z) dz,$$

where $\gamma(t) = re^{-it}$, $t \in [0, 2\pi]$, $r > R$.

△

Notice that γ above is the circle with radius $r > R$ and negative orientation. This is because we consider this circle as a circle about ∞ , i.e. $\gamma(t)$ moves with increasing t anticlockwise with respect to ∞ , clockwise with respect to 0.

There is a connection between $\text{Res}(f, \infty)$ and $\text{Res}(\tilde{f}, 0)$. Notice, however, that in general $\text{Res}(f, \infty) \neq \text{Res}(\tilde{f}, 0)$. This is demonstrated by the exam below.

Example 5.16 Let $f(z) = z$, then $\text{Res}(f, \infty) = \frac{1}{2\pi i} \int_{\gamma} f(z) dz = 0$ since z is holomorphic in $\mathbb{C}$ and γ is a closed path. On the other hand, $\tilde{f}(z) = \frac{1}{z}$, this function has a simple pole at 0. We have $\text{Res}(\tilde{f}, 0) = 1$.

△

Proposition 5.17 Let f be a function holomorphic in $D'(\infty; R)$ with some $R > 0$. Then

$$\text{Res}(f, \infty) = -\text{Res}\left(\frac{\tilde{f}(z)}{z^2}, 0\right).$$

△

Proof. Take $r > R$ and consider the circles $\gamma(t) = re^{-it}$, $t \in [0, 2\pi]$ (negatively oriented) and $\tilde{\gamma}(t) = \frac{1}{\gamma(t)} = \frac{1}{r}e^{it}$, $t \in [0, 2\pi]$ (positively oriented). Notice that

$\tilde{f}(z) = f\left(\frac{1}{z}\right)$ is holomorphic in $D'\left(0; \frac{1}{R}\right)$. We have

$$\begin{aligned} \operatorname{Res}(f, \infty) &= \frac{1}{2\pi i} \int_{\gamma} f(z) dz = \frac{1}{2\pi i} \int_0^{2\pi} f(re^{-it})(-ir)e^{-it} dt \\ &= -\frac{1}{2\pi i} \int_0^{2\pi} f\left(\frac{1}{re^{-it}}\right) \frac{i}{r} e^{it} dt \\ &= -\frac{1}{2\pi i} \int_{\tilde{\gamma}} \frac{\tilde{f}(z)}{z^2} dz = -\operatorname{Res}\left(\frac{\tilde{f}(z)}{z^2}, 0\right). \end{aligned}$$

□

If f has only finitely many singularities in $\overline{\mathbb{C}}$ and γ is a simple closed path that does not pass through any of the singularities, then we can calculate $\int_{\gamma} f(z) dz$ using either the sum of the residues of f at the singularities inside γ , or the sum of the residues of f at the singularities outside γ (taking into account the orientation of γ).

Example 5.18 Consider the function $f(z) = \frac{1}{1+z^{40}}$. This function has 40 simple poles that all lie on the unit circle. Suppose we wish to calculate $\int_{\gamma} f(z) dz$, where $\gamma(t) = 2e^{it}$, $t \in [0, 2\pi]$. Instead of 40 singularities inside γ we can use one singularity at ∞ :

$$\int_{\gamma} f(z) dz = -2\pi i \operatorname{Res}(f, \infty) = 2\pi i \operatorname{Res}\left(\frac{f\left(\frac{1}{z}\right)}{z^2}, 0\right) = 2\pi i \operatorname{Res}\left(\frac{z^{38}}{z^{40} + 1}, 0\right) = 0.$$

△

5.5 Applications to real integrals

Many real integrals can be evaluated using complex analysis. In the section we consider some examples.

Example 5.19 Consider the integral $I = \int_{-\pi}^{\pi} \frac{1}{5+4\cos\theta} d\theta$.

Using the formula $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, we rewrite this integral as

$$I = \int_{-\pi}^{\pi} \frac{1}{5+2(e^{i\theta} + e^{-i\theta})} d\theta = \int_{-\pi}^{\pi} \frac{e^{i\theta}}{2e^{2i\theta} + 5e^{i\theta} + 2} d\theta.$$

Notice that when θ runs from $-\pi$ to π , the term $e^{i\theta}$ traces the unit circle. We parameterize the unit circle as $\gamma(\theta) = e^{i\theta}$, $\theta \in [-\pi, \pi]$. Notice that $\gamma'(\theta) = ie^{i\theta}$. With this parameterization we obtain

$$I = \frac{1}{i} \int_{\gamma} \frac{1}{2z^2 + 5z + 2} dz.$$

The zeros of the denominator $2z^2 + 5z + 2$ are $z_1 = -\frac{1}{2}$ and $z_2 = -2$. The pole $z + 2 = -2$ lies outside the unit circle. Hence,

$$I = 2\pi \operatorname{Res} \left(\frac{1}{2z^2 + 5z + 2}, -\frac{1}{2} \right) = 2\pi \left. \frac{1}{(2z^2 + 5z + 2)'} \right|_{z=-\frac{1}{2}} = 2\pi \left. \frac{1}{4z + 5} \right|_{z=-\frac{1}{2}} = \frac{2\pi}{3}.$$

△

The above method can be applied to many integrals that include sin and cos.

Example 5.20 Consider the integral $I = \int_{-\infty}^{\infty} \frac{1}{x^2 + 2x + 10} dx$.

This is an improper integral, its value is understood as the limit

$$I = \lim_{\substack{A \rightarrow \infty \\ B \rightarrow \infty}} \int_{-B}^A \frac{1}{x^2 + 2x + 10} dx.$$

However, the integrand is positive, and it is easy to see that the integral converges (for example, by comparing it with the converging integral $\int_{-\infty}^{\infty} \frac{1}{(x+1)^2} dx$). Therefore,

$$I = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{x^2 + 2x + 10} dx.$$

Now consider the path γ in $\mathbb{C}$ consisting of the segment $[-R, R]$ and the semicircle parameterized by $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$ (see [Figure 5.2](#)).

The poles of the function $\frac{1}{z^2 + 2z + 10} = \frac{1}{(z+1-3i)(z+1+3i)}$ are $-1 \pm 3i$; only the pole $-1 + 3i$ lies in the upper half-plane. We take $R > |-1 + 3i| = \sqrt{10}$; then this pole lies inside γ . We have

$$\begin{aligned} \int_{\gamma} \frac{1}{z^2 + 2z + 10} dz &= 2\pi i \operatorname{Res} \left(\frac{1}{z^2 + 2z + 10}, -1 + 3i \right) \\ &= 2\pi i \lim_{z \rightarrow -1+3i} \frac{z+1-3i}{(z+1-3i)(z+1+3i)} = 2\pi i \frac{1}{-1+3i+1+3i} = \frac{2\pi i}{6i} = \frac{\pi}{3}. \end{aligned}$$

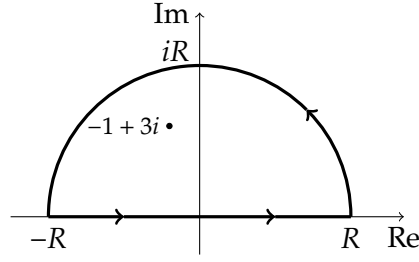


Figure 5.2: Example 5.20

On the other hand,

$$\int_{\gamma} \frac{1}{z^2 + 2z + 10} dz = \int_{[-R, R]} \frac{1}{z^2 + 2z + 10} dz + \int_{\gamma_R} \frac{1}{z^2 + 2z + 10} dz.$$

Let us estimate $\left| \int_{\gamma_R} \frac{1}{z^2 + 2z + 10} dz \right|$. If $|z| = R$, then

$$|z^2 + 2z + 10| \geq |z|^2 - 2|z| - 10 = R^2 - 2R - 10,$$

and for R large enough we have $|z^2 + 2z + 10| \geq \frac{1}{2}R^2$. It follows that

$$\left| \frac{1}{z^2 + 2z + 10} \right| \leq \frac{2}{R^2}.$$

By the Estimation Theorem (Corollary 2.7),

$$\left| \int_{\gamma_R} \frac{1}{z^2 + 2z + 10} dz \right| \leq \frac{2}{R^2} \pi R = \frac{2\pi}{R} \rightarrow 0, \quad R \rightarrow \infty.$$

We conclude that

$$I = \lim_{R \rightarrow \infty} \int_{[-R, R]} \frac{1}{z^2 + 2z + 10} dz = \int_{\gamma} \frac{1}{z^2 + 2z + 10} dz = \frac{\pi}{3}.$$

△

The above method can be applied to any integral of the form $\int_{-\infty}^{\infty} f(x) dx$, where the integrand is a rational function $f(x) = \frac{p(x)}{q(x)}$ with two polynomials p, q such that

- (1) q has no zeros on the real axis,

$$(2) \deg(q) - \deg(p) \geq 2.$$

The first condition ensures that f does not have singularities on the real axis. The second condition guarantees that the integral over the semicircle of radius R tends to zero as $R \rightarrow \infty$.

Sometimes it is possible to apply this method, possibly with some modifications, even if the above conditions are not satisfied. However, the improper integral $\int_{-\infty}^{\infty} f(x) dx$ might not converge, and in general $\int_{-\infty}^{\infty} f(x) dx$ may differ from $\lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx$.

Example 5.21 Consider the integral $\int_{-\infty}^{\infty} \frac{2x}{1+x^2} dx$.

This integral diverges, because

$$\lim_{R \rightarrow \infty} \int_0^R \frac{2x}{1+x^2} dx = \lim_{R \rightarrow \infty} \ln(1+x^2) \Big|_0^R = \infty.$$

However,

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{2x}{1+x^2} dx = 0$$

because $\frac{2x}{1+x^2}$ is an odd function.

△

In some cases it is useful to think of $\lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx$ as a “value” of $\int_{-\infty}^{\infty} f(x) dx$, even if $\int_{-\infty}^{\infty} f(x) dx$ diverges. This idea is formalized in the following definition.

Definition 5.22 (1) The *Cauchy principal value* of the integral $\int_{-\infty}^{\infty} f(x) dx$ is defined as

$$\text{P.V.} \int_{-\infty}^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx.$$

(2) If f has a singularity at $c \in (a, b) \subseteq \mathbb{R}$, then the *Cauchy principal value* of the integral $\int_a^b f(x) dx$ is defined as

$$\text{P.V.} \int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0+} \left(\int_a^{c-\varepsilon} f(x) dx + \int_{c+\varepsilon}^b f(x) dx \right).$$

△

This definition can be extended in an obvious manner to the situation when the integrand f has several isolated singularities on $\mathbb{R} \cup \{\infty\}$.

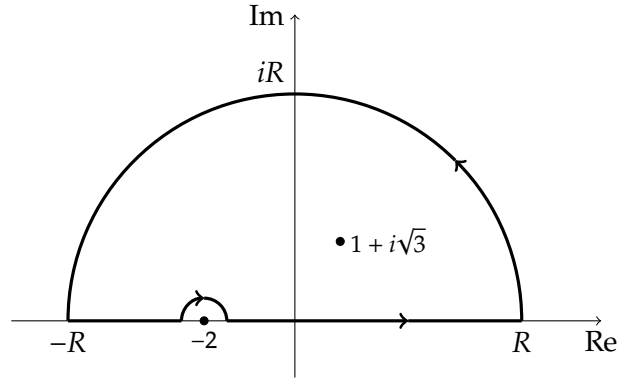


Figure 5.3: Example 5.23

Example 5.23 Let us evaluate P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3+8} dx$.

The function $\frac{1}{z^3+8}$ has three simple poles $-2, 1 \pm i\sqrt{3}$. Choose $R, \varepsilon > 0$ with $R > 2 + \varepsilon$. Consider the path γ shown in Figure 5.3, where the small semicircle about -2 has radius ε .

We have

$$\int_{\gamma} \frac{1}{z^3+8} dz = 2\pi i \operatorname{Res}\left(\frac{1}{z^3+8}, 1+i\sqrt{3}\right) = 2\pi i \left. \frac{1}{3z^2} \right|_{z=1+i\sqrt{3}} = \frac{\pi(\sqrt{3}-i)}{12}.$$

Let $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$. If $|z| = R$, then for R large enough $|z^3+8| \geq |z|^3-8 = R^3-8 \geq \frac{1}{2}R^3$, and

$$\left| \int_{\gamma_R} \frac{1}{z^3+8} dz \right| \leq \frac{2}{R^3} \pi R = \frac{2\pi}{R^2} \rightarrow 0, \quad R \rightarrow \infty.$$

Now consider $\gamma_{\varepsilon}(t) = -2 - \varepsilon e^{-it}$, $t \in [0, \pi]$. By Lemma 5.24 below we have

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\gamma_{\varepsilon}} \frac{1}{z^3+8} dz = -\pi i \operatorname{Res}\left(\frac{1}{z^3+8}, -2\right) = -\pi i \left. \frac{1}{3z^2} \right|_{z=-2} = -\frac{\pi i}{12}.$$

Since

$$\left(\int_{\gamma_R} + \int_{[-R, -2-\varepsilon]} + \int_{\gamma_{\varepsilon}} + \int_{[-2+\varepsilon, R]} \right) \frac{1}{z^3+8} dz = \int_{\gamma} \frac{1}{z^3+8} dz = \frac{\pi(\sqrt{3}-i)}{12},$$

we conclude that

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1}{x^3 + 8} dx = \frac{\pi(\sqrt{3} - i)}{12} + \frac{\pi i}{12} = \frac{\pi\sqrt{3}}{12}.$$

△

Lemma 5.24 *Let f be a holomorphic function in $D'(z_0; r)$ with some $r > 0$, and let z_0 be a simple pole of f . Consider the path $\gamma_\varepsilon(t) = z_0 + \varepsilon e^{it}$, $t \in [\theta_1, \theta_2]$, where $0 < \varepsilon < r$ and $0 \leq \theta_1 < \theta_2 \leq 2\pi$. Then*

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\gamma_\varepsilon} f(z) dz = i(\theta_2 - \theta_1) \text{Res}(f, z_0).$$

△

Proof. Denote $b = \text{Res}(f, z_0)$. By **Theorem 5.12**,

$$b = \lim_{z \rightarrow z_0} (z - z_0)f(z).$$

Put

$$g(z) = (z - z_0)f(z) - b.$$

Given $\eta > 0$, there exists $\delta > 0$ such that $0 < |z - z_0| < \delta \implies |g(z)| < \eta$.

Now let $0 < \varepsilon < \min\{r, \delta\}$. Notice that $\gamma'_\varepsilon(t) = \varepsilon i e^{it} = i(\gamma_\varepsilon(t) - z_0)$. We have

$$\begin{aligned} \left| \int_{\gamma_\varepsilon} f(z) dz - i(\theta_2 - \theta_1)b \right| &= \left| \int_{\theta_1}^{\theta_2} (f(\gamma_\varepsilon(t))\gamma'_\varepsilon(t) - ib) dt \right| \\ &= \left| \int_{\theta_1}^{\theta_2} g(\gamma_\varepsilon(t)) dt \right| < \eta(\theta_2 - \theta_1), \end{aligned}$$

and the statement follows. □

Remark Notice that this formula is, in general, not valid if z_0 is a pole of order > 1 . △

Our last example demonstrates a method for calculating some integrals of the form $\int_{-\infty}^{\infty} f(x) \sin x dx$ or $\int_{-\infty}^{\infty} f(x) \cos x dx$, where f is a rational function.

Example 5.25 Consider the integral $\int_0^\infty \frac{x \sin x}{1+x^4} dx$.

Notice that the integrand is an even function, and thus for any $R > 0$ we have

$$\int_0^R \frac{x \sin x}{1+x^4} dx = \frac{1}{2} \int_{-R}^R \frac{x \sin x}{1+x^4} dx.$$

On the other hand, $\frac{x \cos x}{1+x^4}$ is an odd function, and therefore

$$\int_{-R}^R \frac{x \cos x}{1+x^4} dx = 0.$$

We conclude that

$$\int_0^R \frac{x \sin x}{1+x^4} dx = \frac{1}{2i} \int_{-R}^R \frac{x e^{ix}}{1+x^4} dx.$$

Let γ be the path like in [Figure 5.2](#) consisting of the segment $[-R, R]$ and the semicircle $\gamma_R(t) = R e^{it}$, $t \in [0, \pi]$. The function $\frac{z e^{iz}}{1+z^4}$ has four simple poles. Two of them, $z_1 = \frac{1}{\sqrt{2}}(-1+i)$ and $z_2 = \frac{1}{\sqrt{2}}(1+i)$ lie in the upper half-plane. We choose R large enough so that they lie inside γ . By the Residue Theorem we have

$$\int_\gamma \frac{z e^{iz}}{1+z^4} dz = 2\pi i \left\{ \text{Res} \left(\frac{z e^{iz}}{1+z^4}, \frac{1}{\sqrt{2}}(-1+i) \right) + \text{Res} \left(\frac{z e^{iz}}{1+z^4}, \frac{1}{\sqrt{2}}(1+i) \right) \right\}.$$

The residues can be calculated as follows:

$$\text{Res} \left(\frac{z e^{iz}}{1+z^4}, \frac{1}{\sqrt{2}}(-1+i) \right) = \frac{z e^{iz}}{4z^3} \Big|_{z=\frac{1}{\sqrt{2}}(-1+i)} = \frac{e^{\frac{1}{\sqrt{2}}(-i-1)}}{-4i}$$

and

$$\text{Res} \left(\frac{z e^{iz}}{1+z^4}, \frac{1}{\sqrt{2}}(1+i) \right) = \frac{z e^{iz}}{4z^3} \Big|_{z=\frac{1}{\sqrt{2}}(1+i)} = \frac{e^{\frac{1}{\sqrt{2}}(i-1)}}{4i}.$$

Thus,

$$\begin{aligned} \int_\gamma \frac{z e^{iz}}{1+z^4} dz &= \frac{\pi}{2} \left(e^{\frac{1}{\sqrt{2}}(i-1)} - e^{\frac{1}{\sqrt{2}}(-i-1)} \right) = \frac{\pi}{2} e^{-\frac{1}{\sqrt{2}}} \left(e^{\frac{1}{\sqrt{2}}i} - e^{-\frac{1}{\sqrt{2}}i} \right) \\ &= i\pi e^{-\frac{1}{\sqrt{2}}} \sin \frac{1}{\sqrt{2}}. \end{aligned}$$

Chapter 5 Singularities and residues

Now let us estimate $\left| \int_{\gamma_R} \frac{ze^{iz}}{1+z^4} dz \right|$. It $|z| = R$ and $z = x + iy$ with $y > 0$, then

$$|e^{iz}| = |e^{ix}e^{-y}| \leq 1,$$

and for R large enough

$$|1 + z^4| \geq |z|^4 - 1 = R^4 - 1 \geq \frac{1}{2}R^4.$$

It follows that

$$\left| \frac{ze^{iz}}{1+z^4} \right| \leq \frac{R}{\frac{1}{2}R^4} = \frac{2}{R^3}$$

on γ_R . Hence,

$$\left| \int_{\gamma_R} \frac{ze^{iz}}{1+z^4} dz \right| \leq \frac{2}{R^3} \pi R \rightarrow 0, \quad R \rightarrow \infty.$$

It follows that

$$\int_0^\infty \frac{x \sin x}{1+x^4} dx = \frac{1}{2i} i\pi e^{-\frac{1}{\sqrt{2}}} \sin \frac{1}{\sqrt{2}} = \frac{\pi}{2} e^{-\frac{1}{\sqrt{2}}} \sin \frac{1}{\sqrt{2}}.$$

△

Chapter 6

Laurent series

We know that if f is holomorphic in $D(z_0; r)$ with some $r > 0$, then we can represent f as a power series $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$. However, if f is holomorphic in $D'(z_0; r)$ and z_0 is an isolated singularity of f , we cannot hope for a power series representation like above. Instead we aim for an expansion of the form

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

for $z \in D'(z_0; r)$.

Notice that, by definition, a double-ended series $\sum_{n=-\infty}^{\infty} b_n$ converges if both series $\sum_{n=0}^{\infty} b_n$ and $\sum_{n=1}^{\infty} b_{-n}$ converge. In this case, if $\sum_{n=0}^{\infty} b_n = s^+$ and $\sum_{n=1}^{\infty} b_{-n} = s^-$, then, by definition, the sum of the double-ended series is $\sum_{n=-\infty}^{\infty} b_n = s^+ + s^-$.

For $z_0 \in \mathbb{C}$ and $0 \leq R_1 < R_2 \leq \infty$, we consider the *annulus*

$$A(z_0; R_1, R_2) = \{z \in \mathbb{C} : R_1 < |z - z_0| < R_2\}.$$

We will study functions holomorphic in annuli.

Theorem 6.1 (Laurent decomposition) *Let f be holomorphic in an annulus $A(z_0; R_1, R_2)$. Then f can be represented as*

$$f = f_0 + f_{\infty},$$

where f_0 is holomorphic in $D(z_0; R_2)$, f_{∞} is holomorphic in $D(\infty; R_1)$, and $\lim_{|z| \rightarrow \infty} f_{\infty}(z) = 0$.

△

Proof. Take r_1, r_2 with $R_1 < r_1 < r_2 < R_2$.

First consider $\gamma_2(t) = z_0 + r_2 e^{it}$, $t \in [0, 2\pi]$. Put

$$f_0(z) = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{w - z} dw, \quad z \in D(z_0; r_2).$$

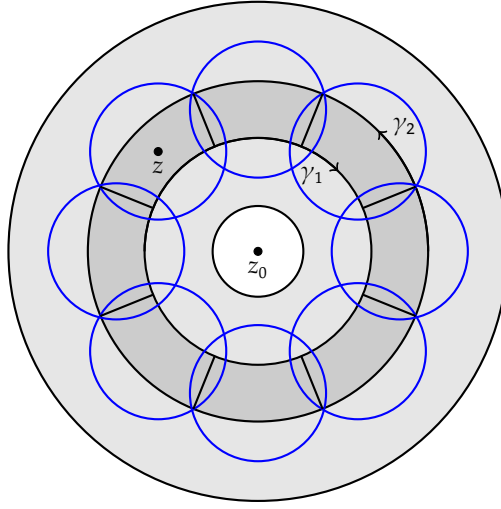


Figure 6.1: Annulus

Repeating the same estimates as in the proof of [Theorem 4.4](#), we can show that the derivative $f'_0(z)$ exists at any point z with $|z| < r_2$. Therefore it is holomorphic in the open disc $D(z_0; r_2)$. Notice that f_0 need not coincide with f for all $z \in A(z_0; R_1, R_2)$.

Now consider the negatively oriented path $\gamma_1(t) = z_0 + r_1 e^{-it}$, $t \in [0, 2\pi]$. Put

$$f_\infty(z) = \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(w)}{w - z} dw, \quad z \in D'(\infty; r_1).$$

Similarly, f_∞ is holomorphic in $D'(\infty; r_1)$. Moreover, $f_\infty(z) \rightarrow 0$ as $|z| \rightarrow \infty$. Thus, f_∞ is holomorphic in $D(\infty; r_1)$, if we extend it by putting $f_\infty(\infty) = 0$. Notice that if f does not have any singularities inside γ_1 , then $f_\infty \equiv 0$ by Cauchy's Integral Theorem. Again, f_∞ need not coincide with f for all $z \in A(z_0; R_1, R_2)$.

The functions f , f_0 and f_∞ are all holomorphic in $A(z_0; r_1, r_2)$. We claim that

$$f(z) = f_0(z) + f_\infty(z), \quad z \in A(z_0; r_1, r_2).$$

To see this, split this annulus into smaller parts such that each of them can be covered by an open disc that is contained in $A(z_0; R_1, R_2)$ (see [Figure 6.1](#)). Apply the Cauchy Integral Formula to the part that contains z , and the Cauchy Integral Theorem to all other parts, and sum up the results.

The same argument shows that the functions f_0 and f_∞ do not depend on particular values of r_1, r_2 . It follows, in particular, that f_0 is holomorphic in $D(z_0; R_2)$, and f_∞ is holomorphic in $D(\infty; R_1)$.

□

Notice that if f does not have any singularities outside γ_2 and is holomorphic at ∞ , then $f_0 \equiv 0$.

Now let us derive a power series representation of f . Without loss of generality, let us assume that $z_0 = 0$.

Since f_0 is holomorphic in $D(0; R_2)$, we can expand it into the Taylor series (see [Theorem 4.7](#))

$$f_0(z) = \sum_{n=0}^{\infty} a_n z^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w^{n+1}} dw$$

and $\gamma(t) = re^{it}$, $t \in [0, 2\pi]$, with any $R_1 < r < R_2$.

To deal with the function f_{∞} , we put

$$\tilde{f}_{\infty}(z) = f_{\infty}\left(\frac{1}{z}\right).$$

Then $\tilde{f}_{\infty}$ is holomorphic in $D\left(0; \frac{1}{R_1}\right)$ and $\tilde{f}_{\infty}(0) = 0$. Its Taylor series is

$$\tilde{f}_{\infty}(z) = \sum_{n=1}^{\infty} b_n z^n$$

(notice that $b_0 = 0$). We can rewrite this as

$$f_{\infty}(z) = \tilde{f}_{\infty}\left(\frac{1}{z}\right) = \sum_{n=1}^{\infty} b_n \left(\frac{1}{z}\right)^n = \sum_{n=-1}^{-\infty} a_n z^n,$$

where $b_n = a_{-n}$. Consider the negatively oriented circle $\tilde{\gamma}(t) = \frac{1}{r}e^{-it}$, $t \in [0, 2\pi]$. Then

$$\begin{aligned} a_{-n} = b_n &= -\frac{1}{2\pi i} \int_{\tilde{\gamma}} \frac{\tilde{f}_{\infty}(z)}{z^{n+1}} dz = -\frac{1}{2\pi i} \int_{\tilde{\gamma}} f_{\infty}\left(\frac{1}{z}\right) \frac{1}{z^{n+1}} dz \\ &= \frac{1}{2\pi i} \int_{\gamma} f_{\infty}(w) w^{n+1} \frac{1}{w^2} dw = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w^{-n+1}} dw. \end{aligned}$$

Altogether we have

$$f(z) = f_0(z) + f_{\infty}(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

We arrive at the following statement.

Theorem 6.2 (Laurent's Theorem) Suppose that f is holomorphic in an annulus $A(z_0; R_1, R_2)$ with $0 \leq R_1 < R_2 \leq \infty$. Then

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw, \quad n \in \mathbb{Z},$$

and $\gamma(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, with any $R_1 < r < R_2$. △

The series $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ is called the *Laurent series* of the function f .

Corollary 6.3 If f is holomorphic in the punctured disc $D'(z_0; r)$ with some $r > 0$ and has in this disc the Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n,$$

then its residue at z_0 is

$$\text{Res}(f, z_0) = a_{-1}.$$
△

This statement follows immediately from the integral formula for a_1 .

Remark Notice that in some books the formula above is taken as the definition of residue. △

The following statement is easy to prove.

Proposition 6.4 Assume that f is holomorphic in the punctured disc $D'(z_0; r)$ with some $r > 0$.

- (1) f has a pole of order m at z_0 if and only if its Laurent series in $D'(z_0; r)$ is of the form

$$f(z) = \sum_{n=-m}^{\infty} a_n(z - z_0)^n, \quad a_{-m} \neq 0.$$

- (2) f has an essential singularity at z_0 if and only if its Laurent series in $D'(z_0; r)$

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

contains infinitely many negative powers of $z - z_0$.

△

Notice that the Laurent series depends on the annulus $A(z_0; R_1, R_2)$. The same function f can have different Laurent series converging in different annuli centered at z_0 . However, if the annulus is fixed, then the Laurent series is unique.

Example 6.5 Consider the function

$$f(z) = \frac{1}{(z-1)(z-3)}.$$

A. First consider the open disc $D(0;1) = \{z \in \mathbb{C} : |z| < 1\}$. The function f is holomorphic in $D(0;1)$ and therefore its Laurent series coincides with its Taylor series. It can be found, for example, as follows:

$$\begin{aligned} f(z) &= \frac{1}{2} \left(\frac{1}{z-3} - \frac{1}{z-1} \right) = \frac{1}{2} \left(-\frac{1}{3} \frac{1}{1-\frac{z}{3}} + \frac{1}{1-z} \right) = \frac{1}{2} \left(-\frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{z}{3} \right)^n + \sum_{n=0}^{\infty} z^n \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{2} \left(1 - \frac{1}{3^{n+1}} \right) z^n. \end{aligned}$$

Notice that the radius of convergence of this series is 1.

B. Now let us derive the Laurent series of f in the annulus $A(0;1,3) = \{z \in \mathbb{C} : 1 < |z| < 3\}$. Its coefficients can be calculated using the integral formulas. Alternatively, the Laurent series can be determined as follows:

$$\begin{aligned} f(z) &= \frac{1}{2} \left(\frac{1}{z-3} - \frac{1}{z-1} \right) = \frac{1}{2} \left(-\frac{1}{3} \frac{1}{1-\frac{z}{3}} - \frac{1}{z} \frac{1}{1-\frac{1}{z}} \right) \\ &= \frac{1}{2} \left(-\frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{z}{3} \right)^n + \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{1}{z} \right)^n \right) \\ &= -\frac{1}{2} \left(\sum_{n=1}^{\infty} z^{-n} + \sum_{n=0}^{\infty} \frac{1}{3^{n+1}} z^n \right). \end{aligned}$$

Notice that this series does not converge if $|z| \leq 1$ or $|z| \geq 3$.

C. Let us determine the Laurent series of f in the disc $D'(\infty;3) = \{z \in \mathbb{C} : |z| > 3\}$.

We have

$$\begin{aligned} f(z) &= \frac{1}{2} \left(\frac{1}{z-3} - \frac{1}{z-1} \right) = \frac{1}{2} \left(\frac{1}{z} \frac{1}{1-\frac{3}{z}} - \frac{1}{z} \frac{1}{1-\frac{1}{z}} \right) \\ &= \frac{1}{2} \left(\frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{3}{z} \right)^n - \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{1}{z} \right)^n \right) = \sum_{n=1}^{\infty} \frac{1}{2} (3^{n-1} - 1) z^{-n}. \end{aligned}$$

D. Finally, let us determine the Laurent series of f in the punctured disc $D'(1;3) = \{z \in \mathbb{C} : 0 < |z-1| < 2\}$. Notice that the pole 1 is the center of this disc, and the pole 3 lies on its boundary.

We have

$$\begin{aligned} f(z) &= \frac{1}{2} \left(\frac{1}{z-3} - \frac{1}{z-1} \right) = \frac{1}{2} \left(\frac{1}{z-1-2} - \frac{1}{z-1} \right) \\ &= \frac{1}{2} \left(-\frac{1}{2} \frac{1}{1-\frac{z-1}{2}} - (z-1)^{-1} \right) = \frac{1}{2} \left(-\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z-1}{2} \right)^n - (z-1)^{-1} \right) \\ &= \sum_{n=0}^{\infty} \frac{(-1)}{2^{n+2}} (z-1)^n - \frac{1}{2} (z-1)^{-1}. \end{aligned}$$

The Laurent series indicates once again that f has a simple pole at 1. Moreover, $\text{Res}(f, 1) = -\frac{1}{2}$.

△

Example 6.6 Consider the function

$$f(z) = \frac{e^{3z}}{(z-1)^5}.$$

Its Laurent series about $z_0 = 1$ is

$$f(z) = (z-1)^{-5} e^3 e^{3(z-1)} = (z-1)^{-5} e^3 \sum_{n=0}^{\infty} \frac{3^n (z-1)^n}{n!} = \sum_{n=-5}^{\infty} \frac{e^3 3^{n+5}}{(n+5)!} (z-1)^n.$$

In particular, f has a pole of order 5 at $z = 1$, and $\text{Res}(f, 1) = \frac{e^3 3^4}{4!} = \frac{27e^3}{8}$.

△

Chapter 7

Arguments

7.1 Complex Logarithms

Recall that for $z \in \mathbb{C} \setminus \{0\}$ the argument of z is defined up to a term $2\pi k$, $k \in \mathbb{Z}$. To emphasize that the argument of z is a set (and not a single number), we will use the bracket notation

$$[\arg z] = \{\theta \in \mathbb{R} : z = |z|e^{i\theta}\}.$$

Notice that this is not a widespread notation. We have

$$[\arg z] = \{\theta + 2\pi k : k \in \mathbb{Z}\},$$

where θ is any fixed number with $z = |z|e^{i\theta}$. For example,

$$[\arg i] = \left\{ \frac{\pi}{2} + 2\pi k : k \in \mathbb{Z} \right\}.$$

To make a well-defined function, one can consider the *principal value* of the argument $\text{Arg } z = \theta$, where $\theta \in (-\pi, \pi]$ and $z = |z|e^{i\theta}$ (sometimes the interval $\theta \in [0, 2\pi)$ is chosen).

Now we are going to define the logarithm of a complex number. We will only consider logarithms to the base e , and we will write for the logarithm $\log$.

Take $z \in \mathbb{C} \setminus \{0\}$. Put $z = e^w = e^{u+iv}$, where $u = \text{Re } w$, $v = \text{Im } w$. Then we have

$$|z| = |e^u||e^{iv}| = e^u,$$

i.e., $u = \text{Re } w = \log |z|$. Moreover,

$$[\arg z] = \{v + 2\pi k : k \in \mathbb{Z}\}.$$

We see that

$$e^w = z \quad \text{if and only if} \quad w = \log |z| + i\theta, \quad \text{where} \quad \theta \in [\arg z].$$

Definition 7.1 For $z \in \mathbb{C} \setminus \{0\}$, we define the *logarithm* of z as the set

$$[\log z] = \{\log |z| + i\theta : \theta \in [\arg z]\}.$$

The *principal value* of the logarithm of z is defined by

$$\text{Log } z = \log |z| + i \text{Arg } z.$$

△

For example, $[\log 2] = \{\log 2 + i2\pi k : k \in \mathbb{Z}\}$, $[\log 1] = \{i2\pi k : k \in \mathbb{Z}\}$, $[\log(-1)] = \{i\pi(2k+1) : k \in \mathbb{Z}\}$, $[\log i] = \left\{i\pi\left(2k + \frac{1}{2}\right) : k \in \mathbb{Z}\right\}$.

Definition 7.2 For $z \in \mathbb{C} \setminus \{0\}$ and $a \in \mathbb{C}$ we define

$$[z^a] = \left\{e^{a(\log |z| + i\theta)} : \theta \in [\arg z]\right\}.$$

△

Notice that the complex power is again defined as a set of numbers. In particular, if $a = \frac{1}{n}$ with $n \in \mathbb{N}$, then the n -th root

$$\left[z^{\frac{1}{n}}\right] = \left\{e^{\frac{1}{n}(\log |z| + i\theta)} : \theta \in [\arg z]\right\}$$

contains exactly n numbers. When $a \in \mathbb{Z}$, then $[z^a]$ contains only one number.

An interesting example is

$$[i^i] = \left\{e^{i\left(i\pi\left(\frac{1}{2} + 2k\right)\right)} : k \in \mathbb{Z}\right\} = \left\{e^{-\frac{\pi}{2}} e^{-2\pi k} : k \in \mathbb{Z}\right\}.$$

Arguments, logarithms and powers are examples of *multifunctions*, i.e. functions that assign to a number $z \in S \subseteq \mathbb{C}$ a finite or infinite subset of $\mathbb{C}$. A function f on $S \subseteq \mathbb{C}$ that assigns to each $z \in S$ a unique complex number $f(z) \in \mathbb{C}$ is called in this context a *single-valued function*.

One could ask the following question: given a multifunction w defined on S , can we select a value $f(z) \in [w(z)]$ such that the single-valued function f is holomorphic in a possibly large open set G ? Consider the logarithm function as an example. If we want to choose $f(z) \in [\log z] = \{\log |z| + i\theta : \theta \in [\arg z]\}$ such that f is continuous on a circle of radius $r > 0$ about 0, then also θ must be continuous. But if z traces a full circle about the origin, the argument of z changes by 2π . We see that we cannot select a holomorphic single-valued function f in an open set that contains the origin.

One way to deal with the problem is to consider a smaller region G chosen in a way that prevents z from tracing a circle about the origin. One can cut the complex plane along a half-line from 0 to ∞ . Usually the cut is made along the negative real axis. The set $\mathbb{C} \setminus (-\infty, 0]$ is an open connected set, i.e. a region. For $k \in \mathbb{Z}$ we put

$$f_k(z) = \log |z| + i(\operatorname{Arg} z + 2\pi k), \quad z \in \mathbb{C} \setminus (-\infty, 0].$$

For each $k \in \mathbb{Z}$, f_k is a single-valued function on $\mathbb{C} \setminus (-\infty, 0]$. The functions f_k are different possible holomorphic logarithms. Such functions are called *branches* of the logarithm, and the point 0 is called the *branch point*.

We have $f'_k(z) = \frac{1}{z}$. To see this, take $z_0, z \in \mathbb{C} \setminus (-\infty, 0]$. Let $w_0 = f_k(z_0)$, $w = f_k(z)$. Then $z_0 = e^{w_0}$, $z = e^w$,

$$f'_k(z_0) = \lim_{z \rightarrow z_0} \frac{f_k(z) - f_k(z_0)}{z - z_0} = \lim_{w \rightarrow w_0} \frac{w - w_0}{e^w - e^{w_0}} = \frac{1}{(e^w)'|_{w=w_0}} = \frac{1}{e^{w_0}} = \frac{1}{z_0}.$$

Another approach is to take infinitely many copies of the complex plane cut along $(-\infty, 0]$ and glue them together at the cut, so that one can pass continuously from one branch of the logarithm to the other when moving to the next copy of the plane. Think of a car park where different levels are connected by ramps. The resulting surface is called the *Riemann surface* of the logarithm.

Similar ideas can be used for other multifunctions.

Consider now an open connected set G on which a holomorphic branch of the logarithm exists. I.e., G must lie in a complex plane cut by a half-line from 0 to ∞ . Let $\theta(z)$ be a continuous function in G with $\theta(z) \in [\arg z]$. Let $f(z) = \log |z| + i\theta(z)$ be a holomorphic branch of the logarithm in G .

Let $\gamma : [a, b] \rightarrow G$ be a path. The net change of the argument when z traces γ from $\gamma(a)$ to $\gamma(b)$ is

$$\theta(\gamma(b)) - \theta(\gamma(a)).$$

On the other hand, by the Fundamental Theorem of Calculus ([Theorem 2.9](#))

$$\int_{\gamma} \frac{dz}{z} = f(\gamma(b)) - f(\gamma(a)) = \log |\gamma(b)| - \log |\gamma(a)| + i(\theta(\gamma(b)) - \theta(\gamma(a))).$$

In particular, if γ is a closed path, then $\int_{\gamma} \frac{dz}{z} = 0$ and $\theta(\gamma(b)) - \theta(\gamma(a)) = 0$.

Now let $\gamma : [a, b] \rightarrow \mathbb{C}$ be an arbitrary closed path in $\mathbb{C}$ with $0 \notin \gamma$. Choose a partition $a = t_0 < t_1 < \dots < t_m = b$ such that every subpath γ_k which is

a restriction of γ to the interval $[t_{k-1}, t_k]$ lies in an open connected set G_k on which a holomorphic branch of the logarithm exists. Let $f_1(z) = \log |z| + i\theta_1(z)$ be an arbitrary holomorphic branch of the logarithm in G_1 . Then determine holomorphic branches $f_k(z) = \log |z| + i\theta_k(z)$ in G_k , $k \geq 2$, such that $f_k(\gamma(t_{k-1})) = f_{k-1}(\gamma(t_{k-1}))$. In this way we obtain a continuous function

$$\theta : [a, b] \rightarrow \mathbb{R}, \quad \theta(t) = \theta_k(t) \quad \text{for } t \in [t_{k-1}, t_k].$$

By above,

$$\int_{\gamma_k} \frac{dz}{z} = \log |\gamma(t_k)| - \log |\gamma(t_{k-1})| + i(\theta(\gamma(t_k)) - \theta(\gamma(t_{k-1}))).$$

When we sum this integrals over all subpaths, the real parts cancel, and we obtain

$$\int_{\gamma} \frac{dz}{z} = \sum_{k=1}^m \int_{\gamma_k} \frac{dz}{z} = i(\theta(b) - \theta(a)).$$

The value $\theta(b) - \theta(a)$ is the net change of the argument when z traces γ , it is of the form $2\pi n$ with $n \in \mathbb{Z}$. The number n is the number of times γ winds around the origin, where anticlockwise windings are counted as positive, and clockwise windings are counted as negative.

Clearly, we can replace 0 by an arbitrary point $z_0 \in \mathbb{C}$. We arrive at the following definition.

Definition 7.3 Let $z_0 \in \mathbb{C}$, and let γ be a closed path in $\mathbb{C}$ with $z_0 \notin \gamma$. The *winding number* (or *index*) of γ about z_0 is the integer

$$n(\gamma, z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z - z_0} dz.$$

△

7.2 Counting zeros and poles

Assume that a function f is holomorphic in a disc $D(z_0; r)$ with some $r > 0$. Recall that in this case f can be represented in $D(z_0; r)$ as the power series

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n, \quad a_n = \frac{f^{(n)}(z_0)}{n!}, \quad n \in \mathbb{N} \cup \{0\}.$$

Definition 7.4 Let f be holomorphic in $D(z_0; r)$, $r > 0$, and not identically zero. If $f(z_0) = 0$, then z_0 is called a **zero** of f . If f has the power series expansion

$$f(z) = \sum_{n=m}^{\infty} a_n(z - z_0)^n, \quad \text{where } m \in \mathbb{N} \text{ and } a_m \neq 0,$$

then z_0 is called a **zero of order** m . Zeros of order 1, 2, ... are called **simple zeros**, **double zeros** and so on. △

The following statement follows immediately from the definition.

Proposition 7.5 Let f be holomorphic in $D(z_0; r)$, $r > 0$, and not identically zero. The following statements are equivalent:

- (1) z_0 is a zero of f of order $m \in \mathbb{N}$,
- (2) $f(z_0) = f'(z_0) = \dots = f^{(m-1)}(z_0) = 0$, $f^{(m)}(z_0) \neq 0$,
- (3) $f(z) = (z - z_0)^m \varphi(z)$, where φ is a holomorphic function in $D(z_0; r)$ and $\varphi(z_0) \neq 0$.

△

The next statement is the central statement of this section.

Theorem 7.6 (The Argument Principle) Let G be a region and γ be a positively oriented closed path in G . Assume that a function f is meromorphic in G and non-zero on γ . Then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = N - P,$$

where N is the number of zeros of f inside γ , P is the number of poles of f inside γ , and zeros and poles of order m are counted m times. △

Proof. The function f is holomorphic in G except at poles and at zeros of f .

Let a be a zero of f of order $m \in \mathbb{N}$. Then

$$f(z) = (z - a)^m \varphi(z),$$

where the function φ is holomorphic and non-zero in some disc $D(a; r)$, $r > 0$. We have

$$f'(z) = m(z - a)^{m-1} \varphi(z) + (z - a)^m \varphi'(z),$$

and consequently

$$\frac{f'(z)}{f(z)} = \frac{m}{z-a} + \frac{\varphi'(z)}{\varphi(z)}.$$

Notice that the function $\frac{\varphi'}{\varphi}$ is holomorphic in $D(a; r)$. It follows that a is a simple pole of $\frac{f'}{f}$ with

$$\text{Res}\left(\frac{f'}{f}, a\right) = m.$$

Now assume that b is a pole of f of order $m \in \mathbb{N}$. Then

$$f(z) = (z-b)^{-m}\varphi(z),$$

where the function φ is holomorphic and non-zero in some disc $D(b; r)$, $r > 0$. We have

$$f'(z) = -m(z-b)^{-m-1}\varphi(z) + (z-b)^{-m}\varphi'(z),$$

and consequently

$$\frac{f'(z)}{f(z)} = \frac{-m}{z-b} + \frac{\varphi'(z)}{\varphi(z)}.$$

As above, $\frac{\varphi'}{\varphi}$ is holomorphic in $D(b; r)$, and a is a simple pole of $\frac{f'}{f}$ with

$$\text{Res}\left(\frac{f'}{f}, b\right) = -m.$$

The result now follows by the Residue Theorem ([Theorem 5.11](#)).

□

Remark By the chain rule,

$$\frac{d}{dz} \log(f(z)) = \frac{f'(z)}{f(z)}.$$

We have seen in the previous section that $\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz$ equals to the net change of the argument, when z traces the curve $f(\gamma)$. This is where the name Argument Principle comes from.

△

The next statement shows that the number of zeros of a holomorphic function in a given region does not change under small perturbations of f .

Theorem 7.7 (Rouche's Theorem) Let G be a region and γ be a positively oriented closed path in G . Assume that f and g are holomorphic in G , and

$$|g(z)| < |f(z)| \quad \text{for all } z \in \gamma.$$

Then f and $f + g$ have the same number of zeros inside γ .

△

Proof. Notice that $|f(z)| > 0$ on γ .

Let $t \in [0, 1]$. Since $|g(z)| < |f(z)|$ on γ , we have $(f + tg)(z) \neq 0$ for all $z \in \gamma$. Consider the function

$$\psi(t) = \frac{1}{2\pi i} \int_{\gamma} \frac{(f' + tg')(z)}{(f + tg)(z)} dz.$$

By the Argument Principle (**Theorem 7.6**), $\psi(t)$ is the number of zeros of the function $f + tg$ inside γ .

We are going to prove that ψ is continuous on $[0, 1]$. Fix $t \in [0, 1]$, take $s \in [0, 1]$, and consider the difference

$$\psi(t) - \psi(s) = \frac{t - s}{2\pi i} \int_{\gamma} \frac{(fg' - f'g)(z)}{(f + tg)(z)(f + sg)(z)} dz.$$

Since all involved functions are continuous on a compact set γ , their moduli are continuous real-valued functions, and they attain their maximum and minimum values on γ . It follows that there exist constants $K, M, m > 0$ such that for all $z \in \gamma$

$$|fg' - f'g|(z) \leq K, \quad |g(z)| \leq M, \quad |(f + tg)(z)| \geq m.$$

We have

$$|(f + sg)(z)| = |(f + tg + (s - t)g)(z)| \geq |(f + tg)(z)| - |s - t||g(z)|.$$

If $|s - t| < \frac{m}{2M}$, then

$$|(f + sg)(z)| > m - \frac{m}{2M}M = \frac{m}{2}, \quad z \in \gamma.$$

By the Estimation Theorem (**Corollary 2.7**),

$$|\psi(t) - \psi(s)| < \frac{|t - s| K^2}{2\pi m^2} \ell(\gamma) \rightarrow 0 \quad \text{as } s \rightarrow t.$$

Therefore, ψ is continuous on $[0, 1]$.

Recall that the function ψ can take only integer values (it gives the number of zeros of the function $f + tg$ inside γ). An integer-valued continuous function must be constant. In particular, $\psi(0) = \psi(1)$. This is exactly the statement that f and $f + g$ have the same number of zeros inside γ .

□

Example 7.8 Consider the equation

$$e^z + z^3 = 0.$$

A. This equation has no solutions with $|z| < \frac{1}{2}$. To see this, take $f(z) = e^z$ and $g(z) = z^3$. The function $f(z) = e^z$ has no zeros. If $|z| = \frac{1}{2}$, then $|g(z)| = |z|^3 = \frac{1}{8}$. On the other hand, on the circle $|z| = \frac{1}{2}$, z can be written in the form $z = \frac{1}{2}(\cos \theta + i \sin \theta)$ with $\theta \in \mathbb{R}$, and

$$|f(z)| = \left| e^{\frac{1}{2}(\cos \theta + i \sin \theta)} \right| = \left| e^{\frac{1}{2} \cos \theta} \right| \geq e^{-\frac{1}{2}} = 0.6065 \dots > \frac{1}{8}.$$

By Rouché's Theorem, $f + g = e^z + z^3$ has no zeros inside the circle $|z| = \frac{1}{2}$.

B. The equation has three solutions inside the circle $|z| < 2$. Now take $f(z) = z^3$ and $g(z) = e^z$. If $|z| = 2$, then $|f(z)| = 8$, but

$$|g(z)| = \left| e^{2(\cos \theta + i \sin \theta)} \right| = \left| e^{2 \cos \theta} \right| \leq e^2 = 7.3890 \dots < 8.$$

The function $f(z) = z^3$ has three zeros inside the circle $|z| = 2$ (namely, the triple zero at 0). It follows that $f + g = e^z + z^3$ has three zeros inside the circle $|z| < 2$.

△

Example 7.9 The polynomial

$$p(z) = z^5 - 6z^4 - z^3 + 3$$

has exactly one zero outside the unit circle.

To see this, take $f(z) = -6z^4$ and $g(z) = z^5 - z^3 + 3$. Then for $|z| = 1$ we have

$$|g(z)| = |z^5 - z^3 + 3| \leq |z|^5 + |z|^3 + 3 = 5 < 6 = |f(z)|.$$

The function $f(z) = -6z^4$ has exactly four zeros inside the unit circle. It follows that $p = f + g$ has exactly four zeros inside the unit circle. The inequality above

also implies that $p = f + g$ does not have zeros at the unit circle. Since p is a polynomial of degree 5, it must have five zeros. We conclude that p has exactly one zero outside the unit circle.

Notice that this zero must be real — this follows from the fact that non-real zeros of a polynomial with real coefficients occur in conjugate pairs.

△

7.3 The open mapping theorem

Theorem 7.10 (*The open mapping theorem*) *Let f be a non-constant holomorphic function in a region $G \subseteq \mathbb{C}$. If $U \subseteq G$ is open, then $f(U)$ is open.*

△

Proof. Let $w_0 \in f(U)$. We have to show that there is an open disc centered at w_0 that lies in $f(U)$.

Take $z_0 \in U$ with $f(z_0) = w_0$. Since U is open, there is $R > 0$ such that the open disc $D(z_0; R) \subseteq U$. Consider the function $g : D(z_0; R) \rightarrow \mathbb{C}$,

$$g(z) = f(z) - w_0.$$

The function g is holomorphic on $D(z_0; R)$ with $g(z_0) = 0$.

Notice that there exists $r_1 > 0$ such that $g(z) \neq 0$ for all $z \in D'(z_0; r_1)$. This follows by the Identity Theorem (**Theorem 4.15**): If each neighborhood of z_0 would contain a zero of g , then the set of zeros would have a limit point in $D(z_0; R)$, and we would have $g(z) = f(z) - w_0 \equiv 0$, i.e. f would be a constant function.

Take $0 < r < r_1$. The function g does not vanish on the circle $|z| = r$. Since $|g|$ is continuous and the circle is compact, $|g|$ attains its minimum value on the circle. It follows that

$$\delta = \min_{|z|=r} |g(z)| > 0.$$

We claim that $D(w_0; \delta) \subseteq f(U)$. To see this, fix $w \in D(w_0; \delta)$ and define the function $h : D(z_0; R) \rightarrow \mathbb{C}$,

$$h(z) = f(z) - w.$$

We have

$$h(z) = g(z) - (w - w_0),$$

and

$$|w - w_0| < \delta \leq |g(z)| \quad \text{on the circle } |z| = r.$$

Chapter 7 Arguments

By Rouché's Theorem (**Theorem 7.7**), g and h have the same number of zeros inside the circle $|z| = r$. But g has at least one zero z_0 . It follows that h has at least one zero, i.e. there is $z \in D(z_0; r) \subseteq U$ with $f(z) = w$. Therefore, $w \in f(U)$, and the statement follows. □

The above theorem says that holomorphic functions map open sets to open sets. Mapping with this property are called *open mappings*.

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